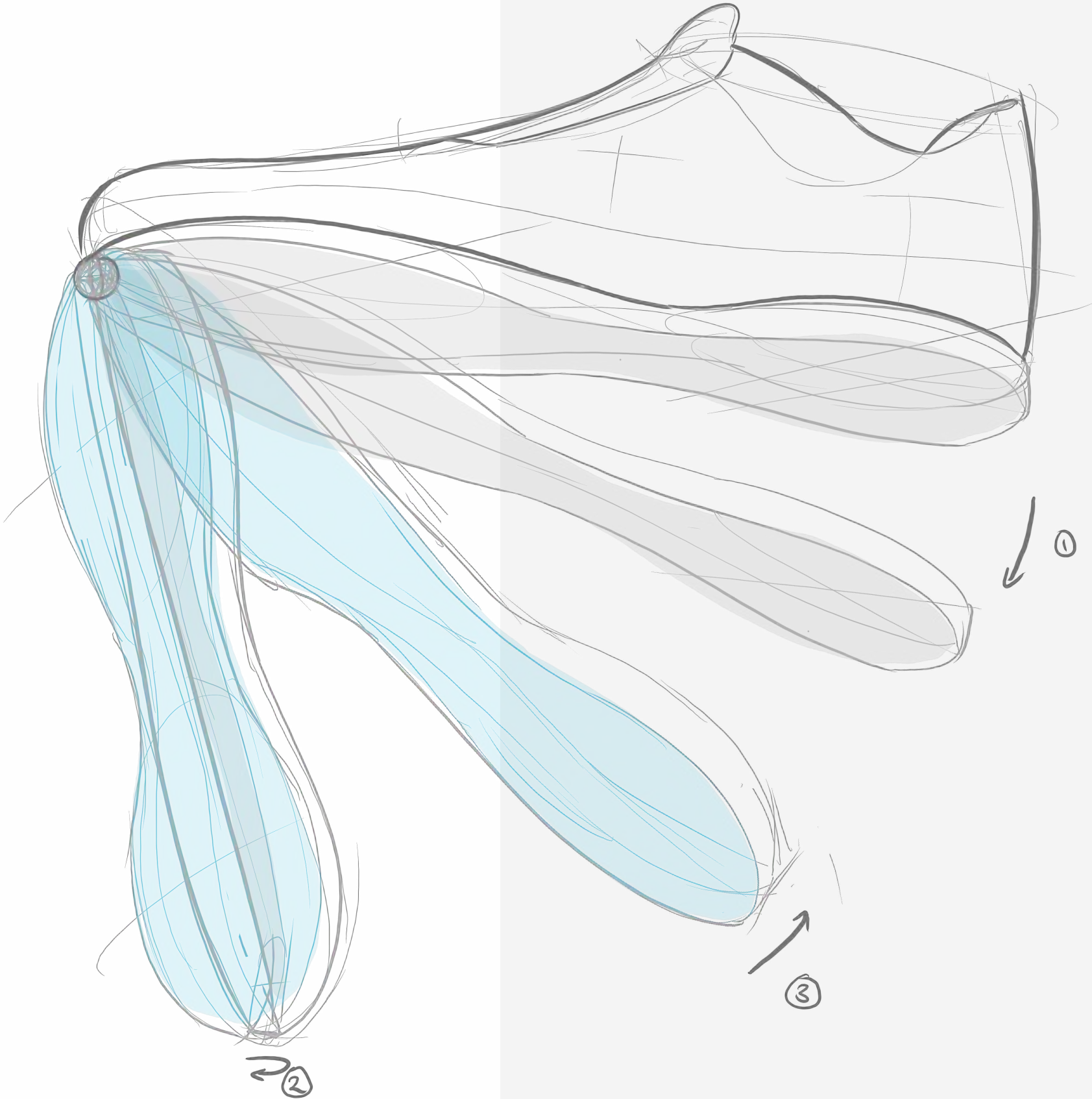


Bisole



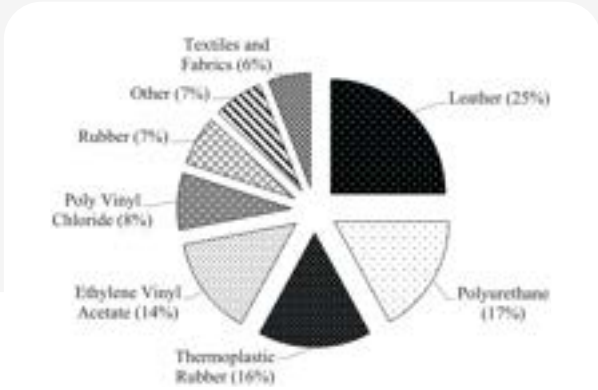
Initial Exploration

‘Transformable Footwear for multiple uses.’

Using this context I researched the possible avenues of design by dividing the broader context up into more managable contexts in which transformable footwear could be useful. After this I conducted research into areas that I found interesting such as environmental issues.



The Environmental Impact of the footwear industry and the shoes created by it are extremely damaging: with an estimated 80% of the waste generated being landfilled***, it's important that the amount of post-use refuse is dramatically reduced. The use of materials with high end of life impacts such as PVC, PU and Vulcanised Rubber are commonly used in soles and making them one of the most harmful footwear components. Leather, which makes up a quarter of the average shoe, is chrome-tanned, which consumes large amounts of water and releases high quantities of dangerous pollutants.



Graph showing the material composition of an average shoe by weight%.

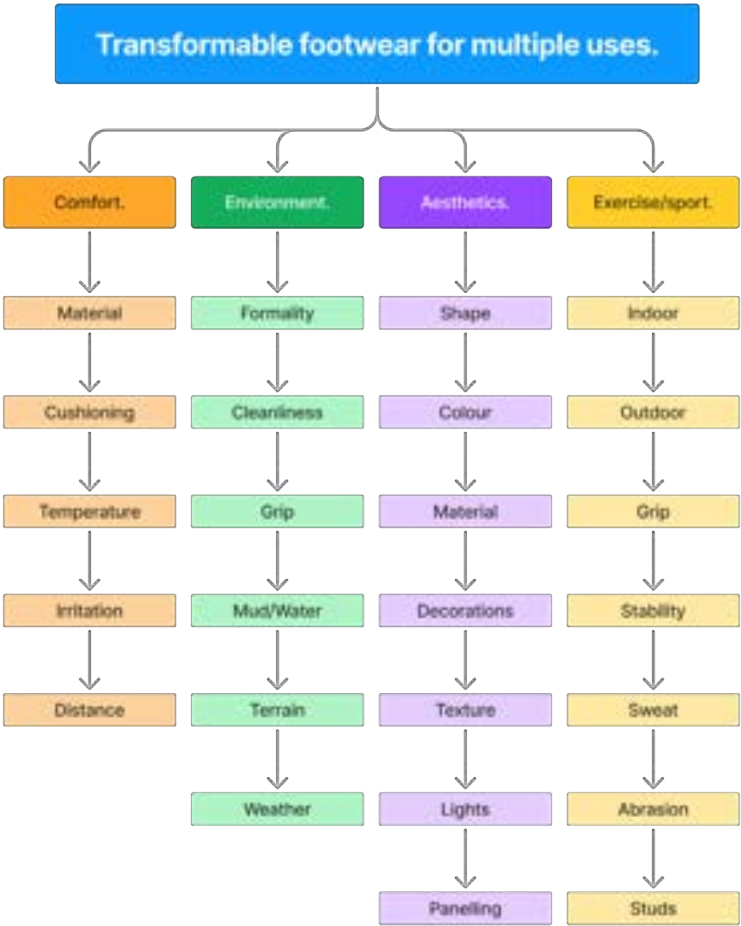
Graph Source: Rahimifard S, Stalkos T, Coates G (2007) Recycling of Footwear Products.

*23.9Bn pairs produced in 2022. Source: Statista Research Department (2024) Amount of footwear produced worldwide from 2015 to 2022. Referencing World Footwear Yearbook 2023.

**Source: Van Rensburg ML, Nkomo SL, Mkhize NM. Life cycle and End-of-Life management options in the footwear industry: A review. Waste Management & Research. 2020;38(6):599-613. doi:10.1177/0734242X20908938

***Source: European Commission LIFE public database. Demonstration of a new business and consumption model for the circular economy in the footwear sector. LIFE19 ENV/ES/000118

****Source: Fiona Armstrong-Gibbs UK Parliament written evidence. Evidence on Sustainability of the fashion industry: the footwear perspective. SF1002



Existing Products:

All shoes shown below with orange accents are currently available to consumers.

Adidas Adifom Q
Lifestyle or fashion shoe that is composed of 2 main parts, the sole and 'exoskeleton' structure as well as the inner lining. Lets the user swap around parts and colour options.

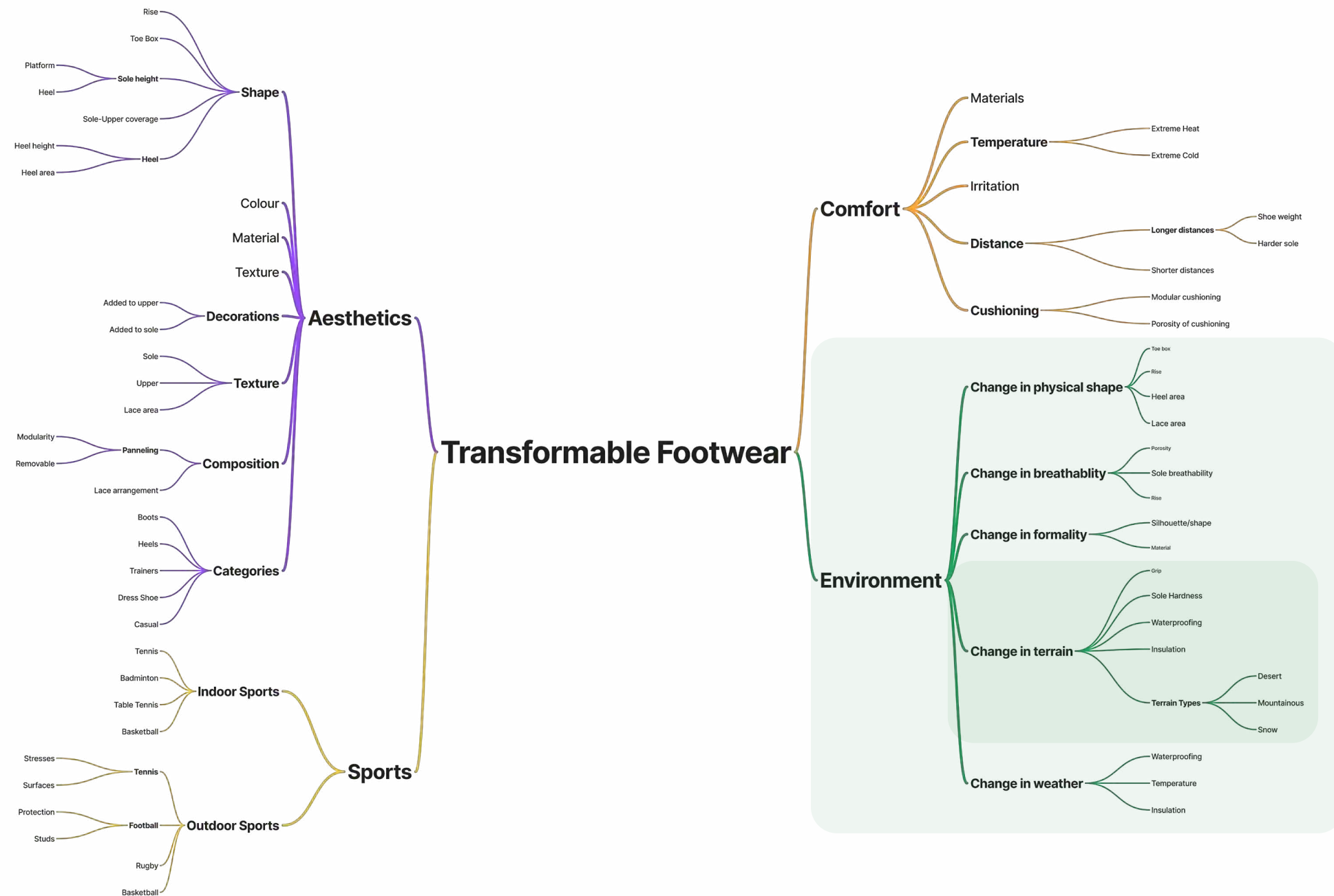
Salomon Snowlog Adv.
Designed for extreme winter conditions, this piece of footwear has soft inner shoe designed for warmth with an outer 'gaiter' that protects against the harsh environment.

Speedland SL:PDX
A performance orientated off road running shoe. The pdx features 'modular' elements such as a removable carbon plate and outsole tread lugs that can be cut into smaller sizes for different terrains.

Crocs and 'Jibbitz'
Although not transformable in a traditional way, the modularity and customisation of jibbitz allows crocs to be easy to personalise and set apart aesthetically.

Group Mindmap

‘Transformable Footwear.’



Group Mindmapping was used after the selection of the design context. It helped my group and I identify the areas within transformable footwear that offered the most interest in further development.

‘Environment’

After the group mindmapping discussion I decided to move forward within the Environment niche for my project. Mainly choosing due to the wider variety of reasons that a shoe can ‘transform’ to its environment.

Within this category, I was particularly interested in changes in the formality of the environment as well as the physical terrain that the footwear is in.

‘Change in Terrain’

Looking further into the category of footwear that transforms to better fit its environment, I opted to select ‘Change in Terrain’. My primary reasoning being that it allows for a wide range of causes for a transformation.

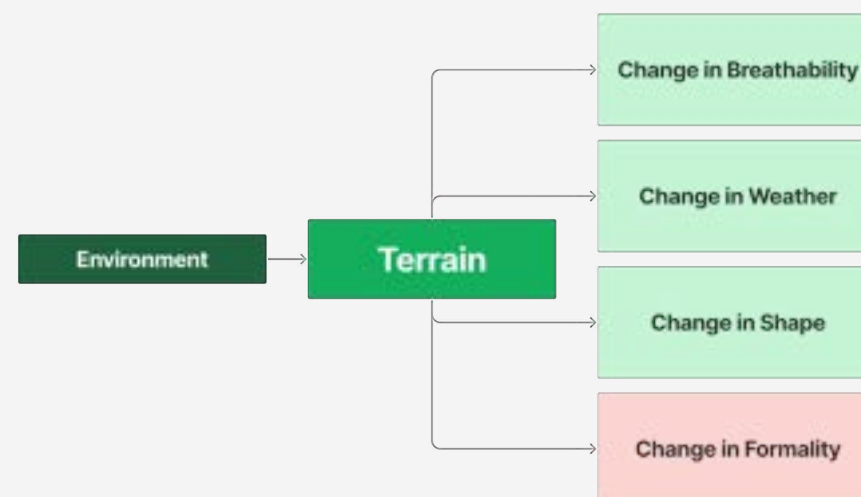
This includes some of the other subcategories that were listed on the group mindmap such as: Changes in Weather or Changes in Breathability.

6+ Pairs

Reccomended for enthusiast or proffesional trail running.*

Specifically looking at sports within the category of ‘Changes in Terrain’, I found that many more terrain intense sports force a user into having multiple pairs of footwear.

This is especially true of trail-running, which expects a user to have around **2-4 pairs** for training on the road and **2-3** which are more suited to the harsher conditions of mountain trails.* Other examples of sports that require a ‘rotation’ of footwear include Tennis for each of the 3 main surfaces and Football for each ground condition archetype.



Choosing Terrain allows me to work within many of the other subcategories except for changes in formality.

‘Transformable footwear for changes in terrain.’

RESERVE

OUTER SHELL

DOWNSTROKE

BENDING TO FORWARD

THINNER CONTACT AREA

evolv

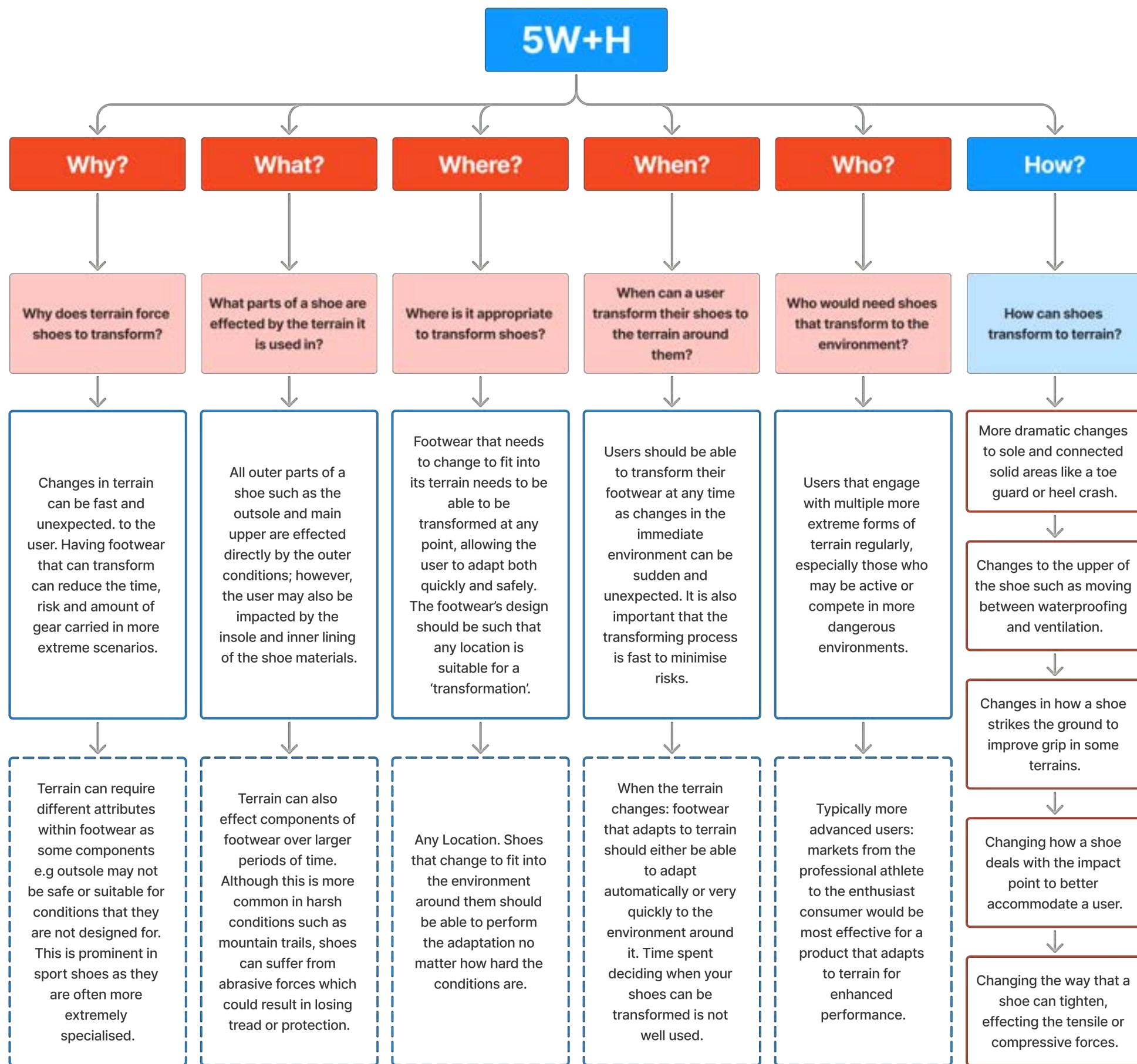
These are an example of footwear that is hyper-specialised to the terrain. They have unique form and features such as an aggressive downturn.

Features are present in different types of footwear. These can be applied or changed to better adapt between shoe 'types' and gain the benefits of both 'types'.

Different concentrations and weaves of mesh are used to change porosity/breathability.

Initial Analysis

5W+1H



In order to generate Insights into the context of transformable footwear I used 5W and 1H questioning. After I established the initial questions for each of the 5Ws I decided to answer each question twice, first when I created the questions then slightly later into the project. This helped to generate insights as I could find common themes between both answers.

Initial Insights

INS (1): typical users are likely to be mid-high athletic performers to require shoes that fit different terrains.

INS (2): having footwear that does not fit the terrain can be dangerous and lead to injury.

INS (3): The faster a shoe can transform the more beneficial

(2): HMW enable footwear to adapt to reduce injury?

(3): HMW allow footwear to change in response to its enviroment?

(1): HMW adapt the properties of shoes to better suit a high perfomance user?

(2): HMW design shoes that change to ensure safety in dangerous terrains?

Generating HMWs

HMW Design **Sports** Footwear to Transform for Different **Terrains** and Activities?

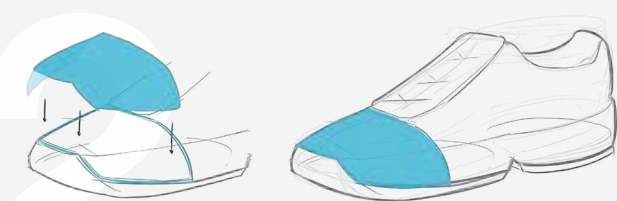
15 Initial Concepts

Idea Generation

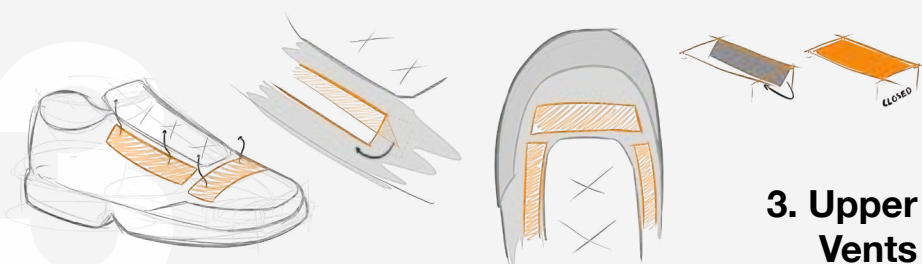
Using my initial and secondary Ideation as well as research conducted in the first section of this project, I created fifteen individual concepts before narrowing my project down to three and eventually a single finalised concept.



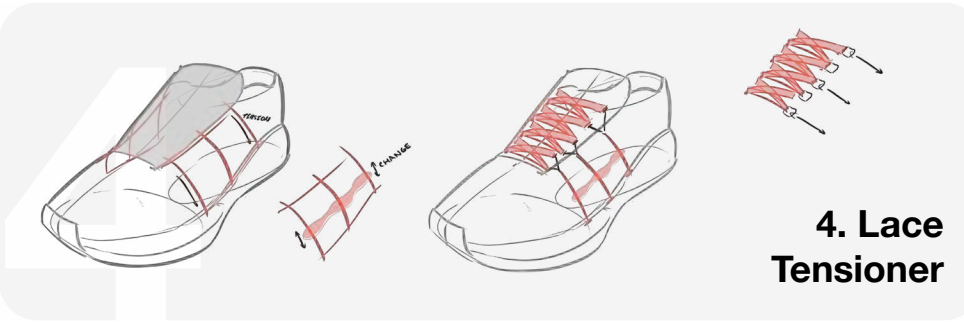
1. Folding Bootguard



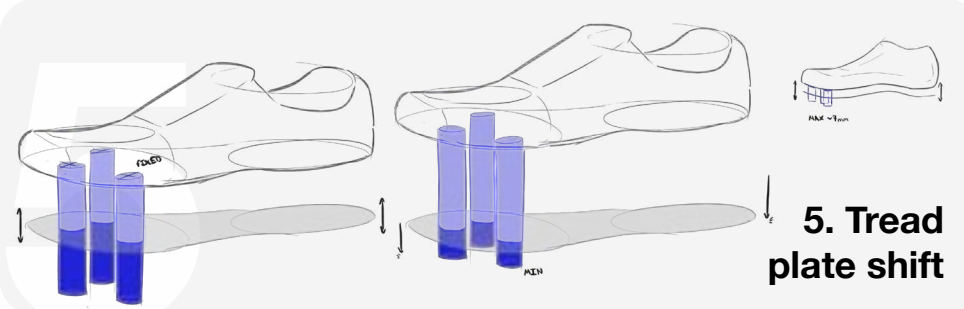
2. Clip on Toeguard



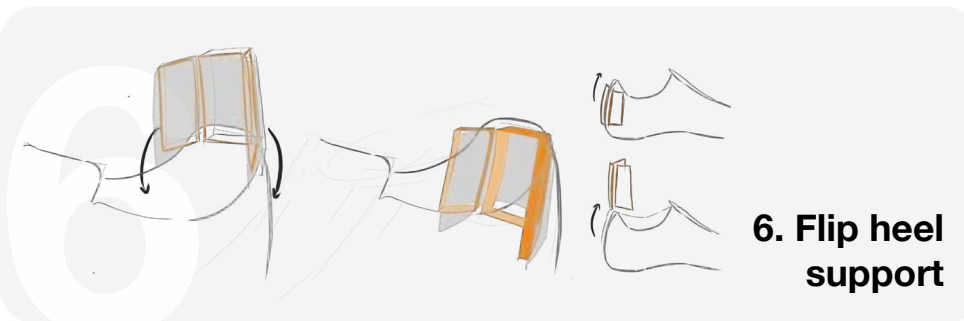
3. Upper Vents



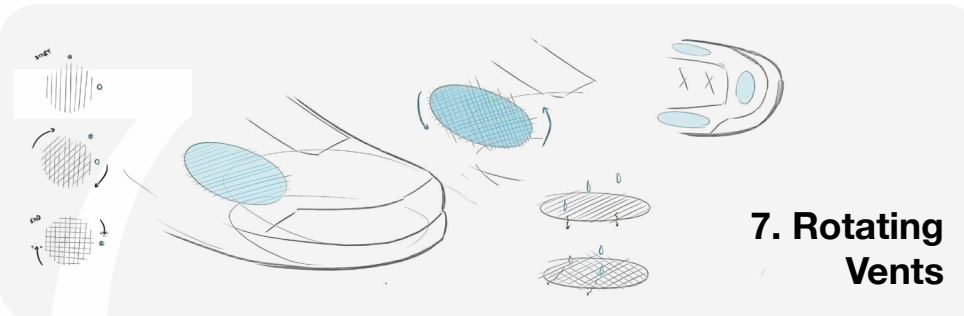
4. Lace Tensioner



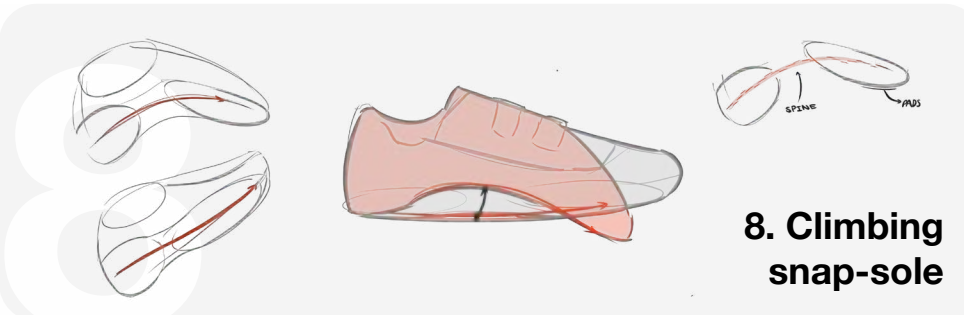
5. Tread plate shift



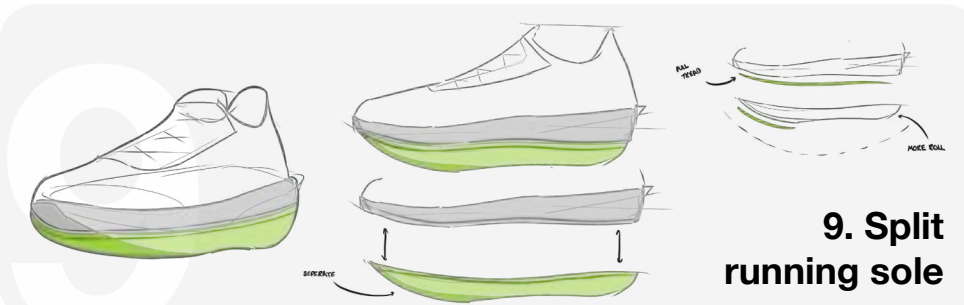
6. Flip heel support



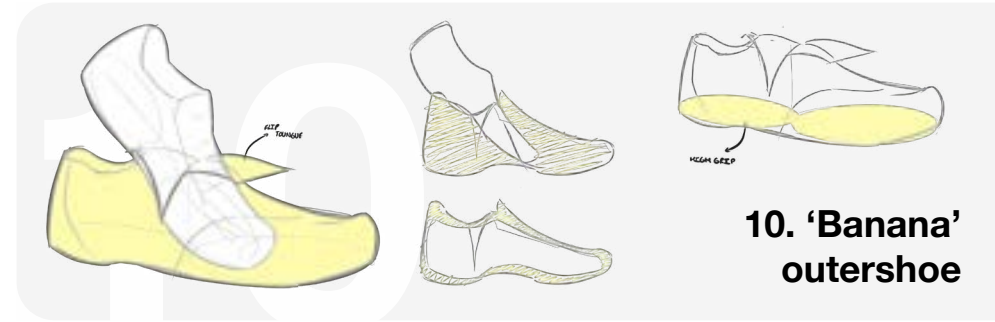
7. Rotating Vents



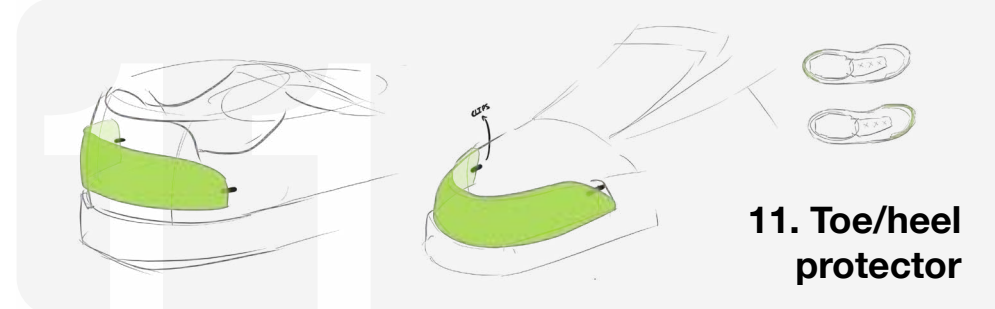
8. Climbing snap-sole



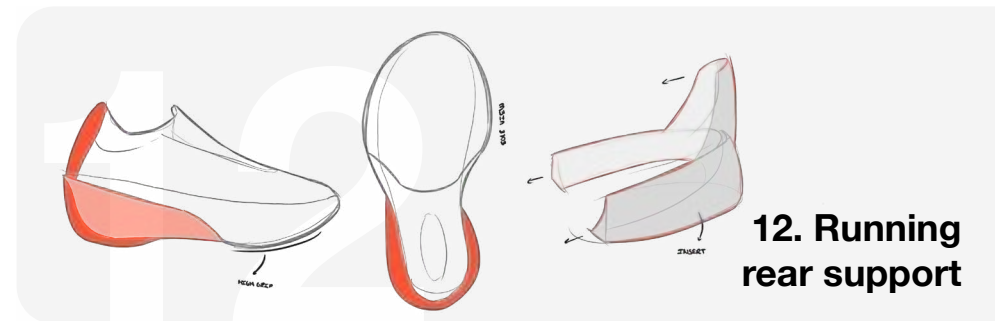
9. Split running sole



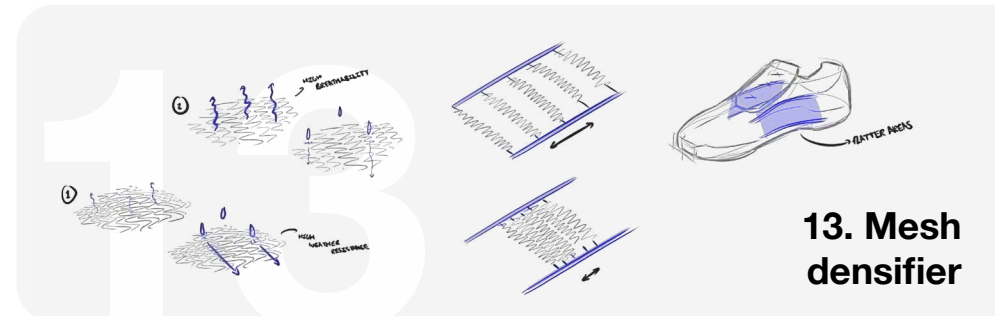
10. 'Banana' outershoe



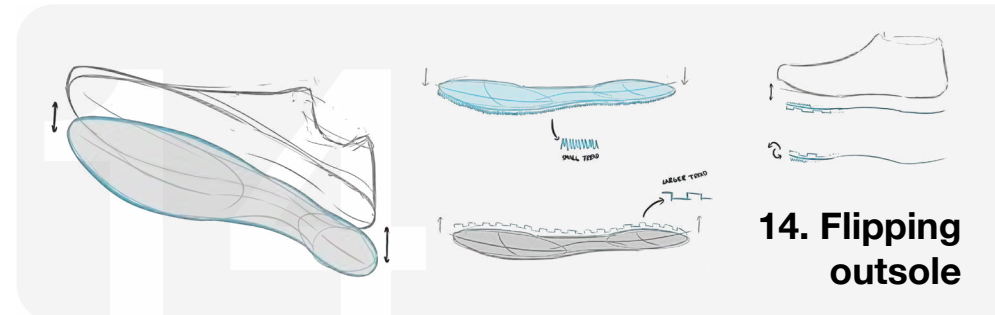
11. Toe/heel protector



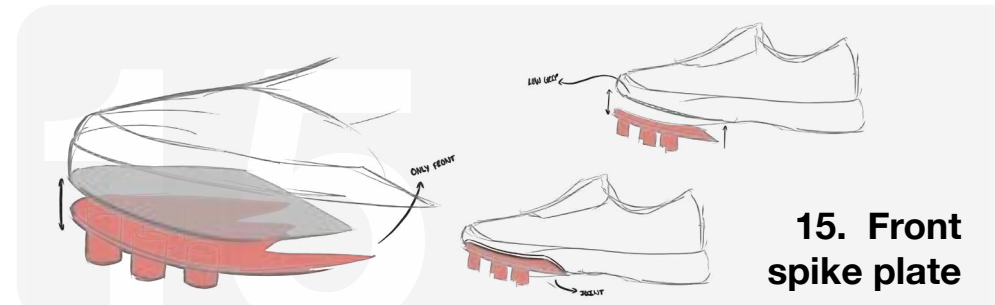
12. Running rear support



13. Mesh densifier



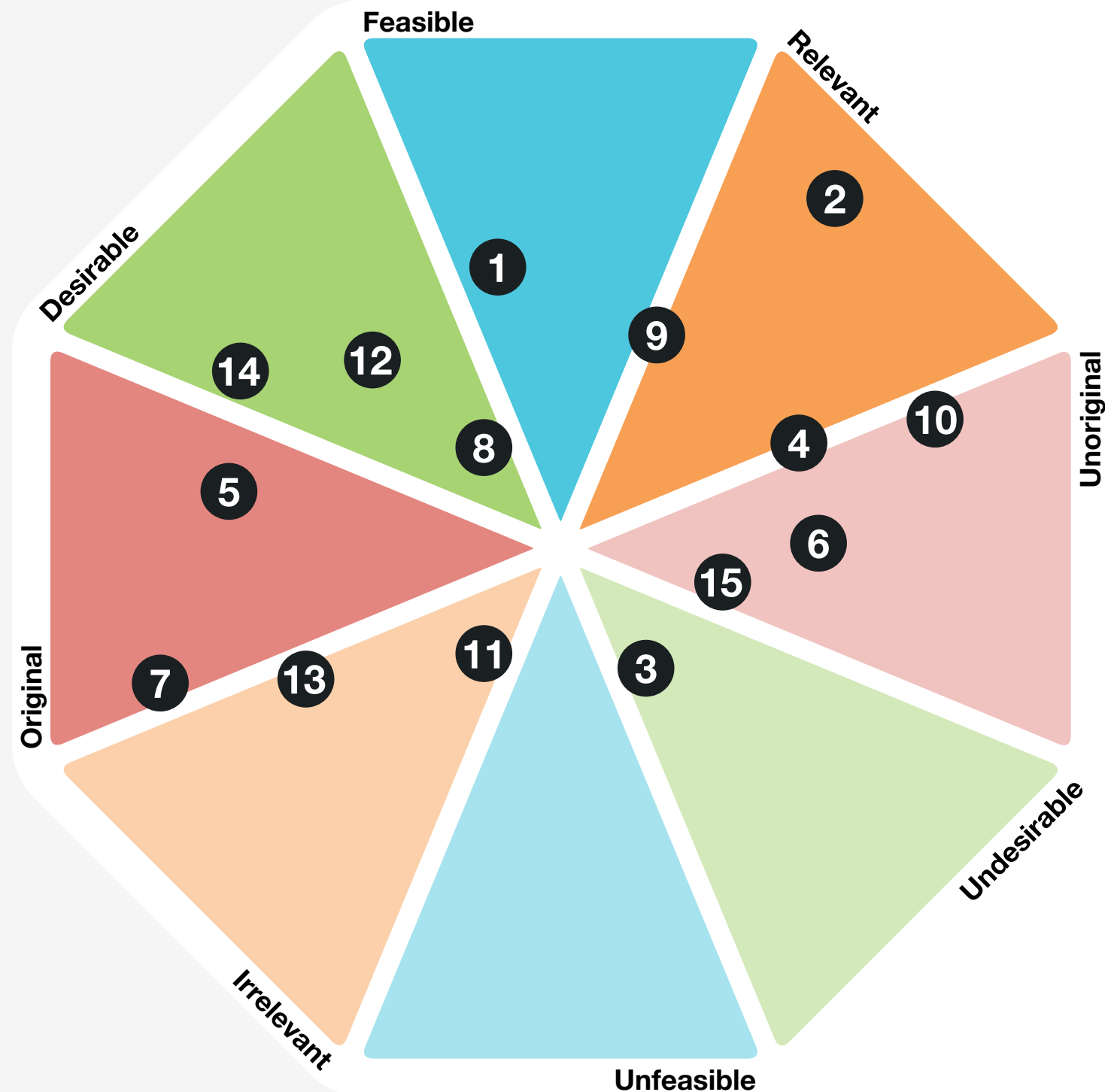
14. Flipping outsole



15. Front spike plate

Multi-Axis plot:

Starting my analysis I chose to graph all of my 15 initial concepts on four categories: *Originality* of the idea; *Desirability* of the product; *Feasibility* of execution and *Relevancy* to the original category.



ANALYSIS: Multi-Axis Plot

Through Radar or Multi-Axis plotting I found that ideas 1, 7 and 14 stood out as having positive qualities on at least 3 scales.

I also selected these concepts as they had a high value for originality.

1
7
14

ANALYSIS: Peer Review

After asking my peers to choose the best Ideas based on the same categories, I found that Ideas 2, 4, 7, 9 and 14 were popular.

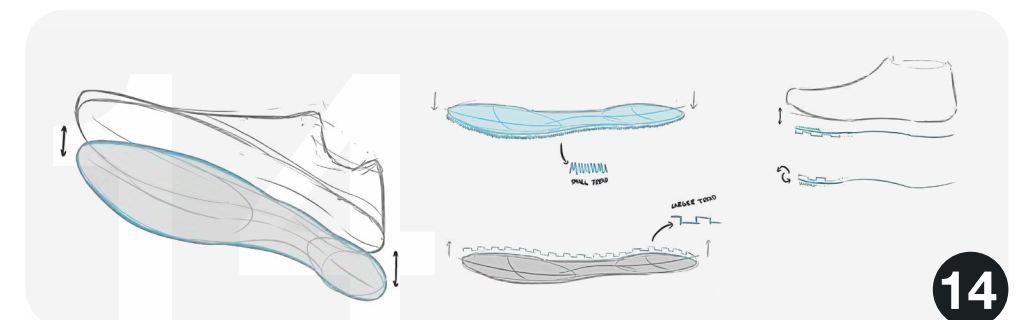
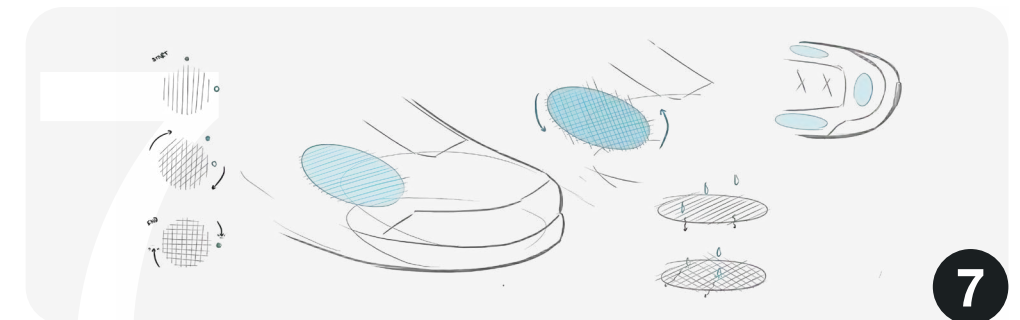
This exercise also helped me to identify concepts that were not clearly presented.

2
4
7
9
14

15 Ideas Analysis

Using a Multi-Axis Plot and Peer Review

In order to select the three best concepts from my initial fifteen, I used a multi axis plot to visualise how all of the ideas compared to each other. Additionally, I used peer review for my fifteen concepts in order to get differing opinions on which ideas to take forward.



Lo-Fi

Prototypes

Modelling my 3 Chosen Concepts.

In order to further analyse the feasibility and function of my three chosen concepts, I used low-fidelity prototypes. These showed me a lot of the physical problems and limitations of the designs and allowed me to explore ways to improve.

Prototyping Process:



1 Folding Bootguard



While prototyping this concept, I discovered that the concept was good overall but would require a much more flexible material to get a tight fit over the sole area. Ideally, it would also be better shaped to the upper of the shoe. This would prevent issues with the guard protruding from over the outsole.

7 Rotating Vents



In my Lo-fi prototyping, I wanted to compare using a 2 mesh layers versus lines in single directions, finding that the single lines provided more variance when aligned and not aligned. I also tested the flexibility of the double layer mesh as well as looking into a centre vs circumference fixing system.

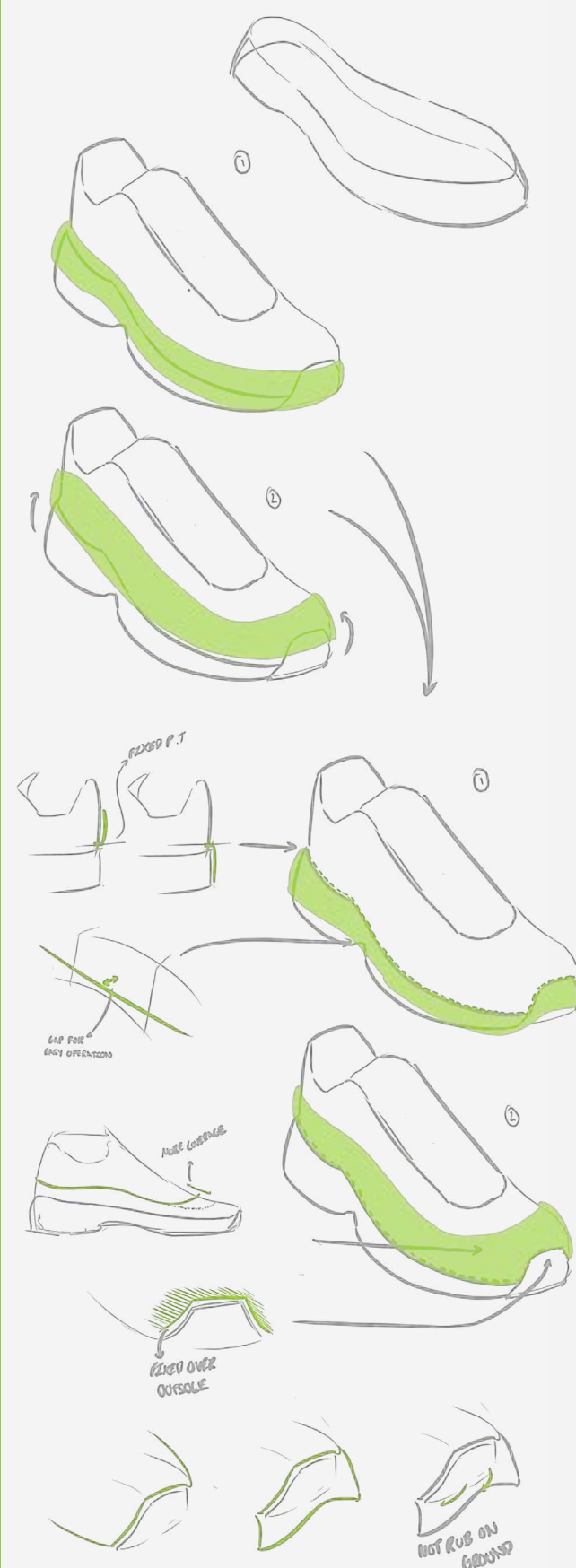
14 Flipping Outsole



Through prototyping, I found that the concept was feasible, especially with the mirrored profiles on each face of the outsole. However, there were some problems with the symmetry of the sole's shape.

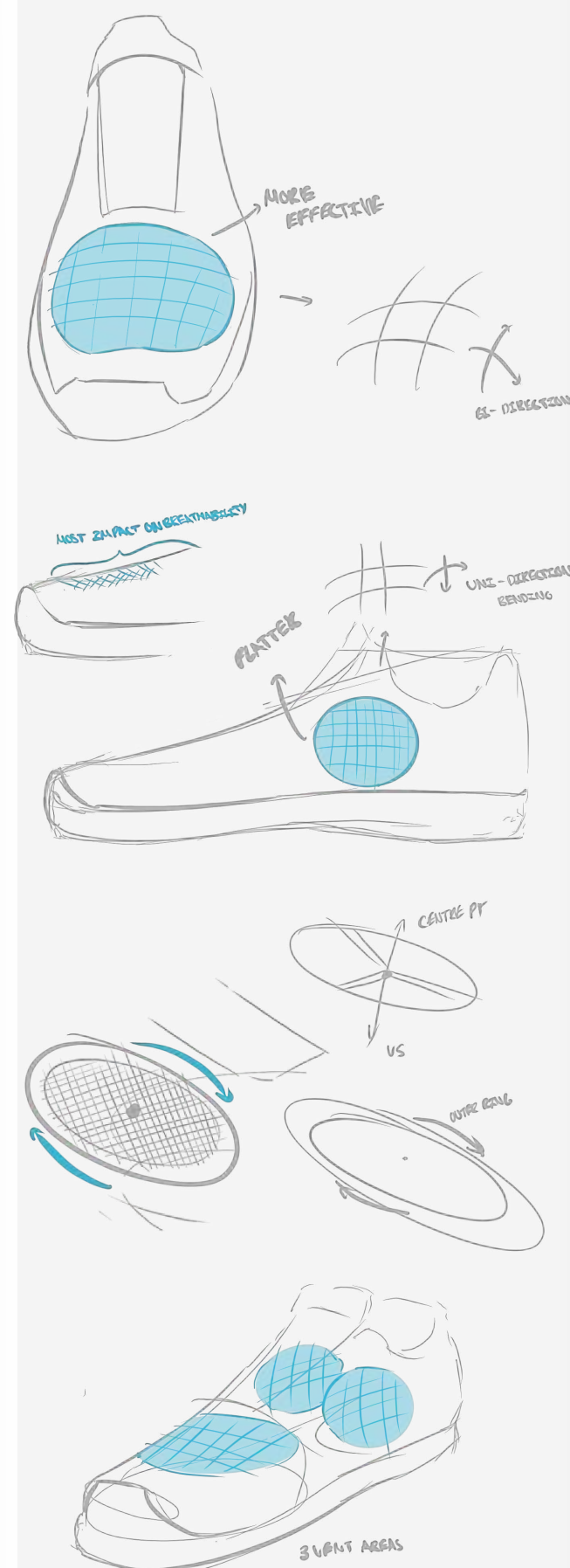
Folding Bootguard

1



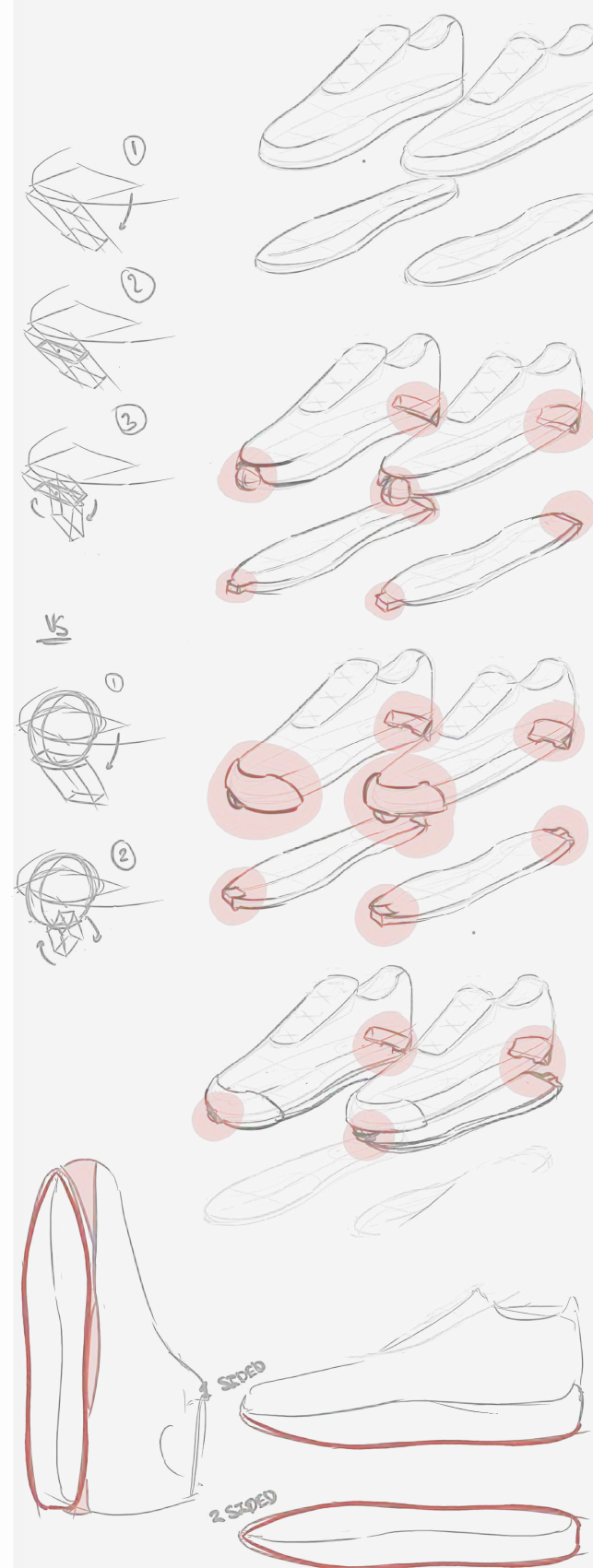
Rotating Vents

7



Flipping Outsole

14



Three Concepts

Further Idea Development

After my Lo-Fi prototyping, I took forward my three selected concepts and further developed my Ideas. This allowed me to get a better understanding of the mechanisms and interactions featured in my shoe concepts.



Analysing the Originality, Feasibility and Desirability of the concepts for a second time, after my further development.

Single Concept

And Design Aim/Brief

After the prototyping stage, I chose to continue with my flipping outsole or 'BiSole' concept. My main reasoning being that this idea was more mechanically challenging and unique than my other 2 selected concepts.

Design Brief/Aim

HMW: Design Sports Footwear to Transform for Different Terrains and Activities?

It is currently well accepted that within running, trail running and some other athletic disciplines, a 'rotation' of shoes is necessary. Not only does this contribute to the environmental impact of the footwear industry, it also creates a barrier which in turn makes these sports harder to access.

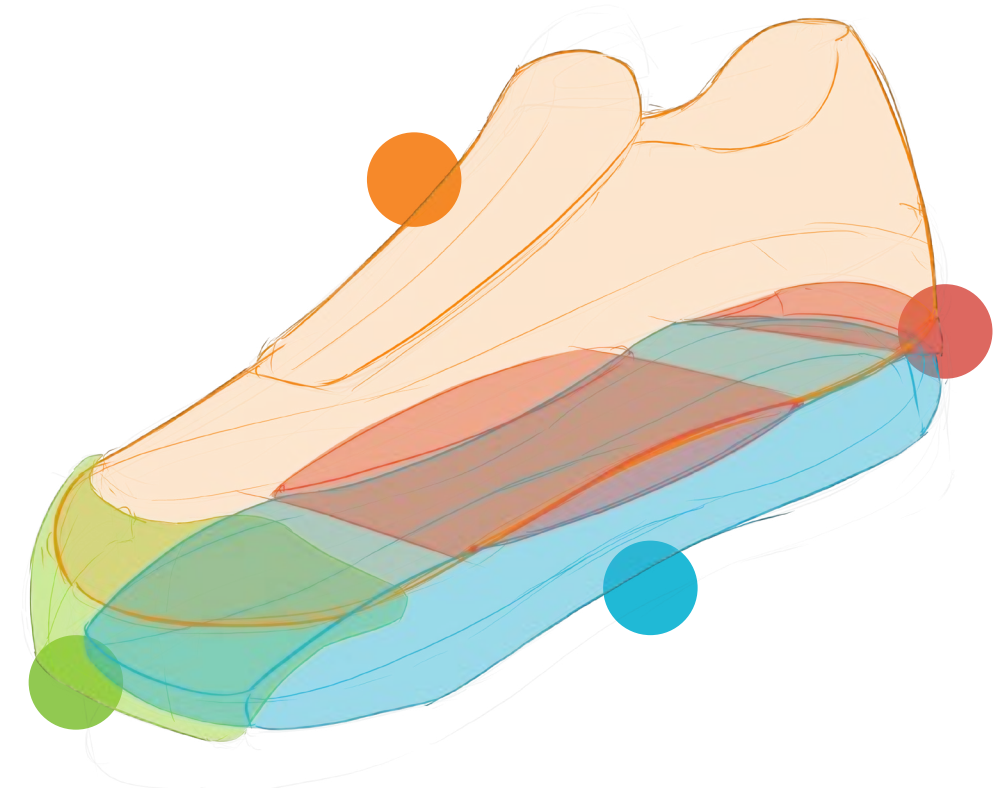
The BiSole aims to reduce the need for so many pairs of shoes. With a single pair being able to adapt and change to perform under a multitude of terrain conditions.

In order to be successful, the BiSole product should target a younger (18-35) age group and be focused on an 'enthusiast' or 'athlete' user category.



Requirements And Design Specification

After the selection of my single idea, I broke down the BiSole concept into it's main physical features. This allowed me to come up with the requirements that are necessary for each element of the shoe.



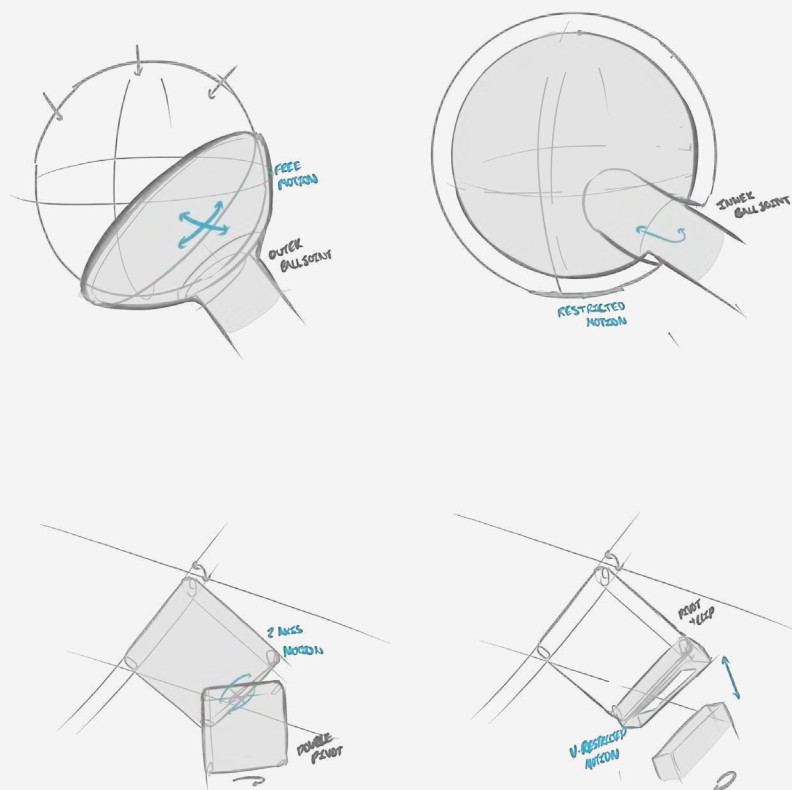
No.	Requirements:	Specifications:
1	The closing of the BiSole must be reliable and resist to harsh terrains and weather conditions.	The closing of the BiSole must allow for 3mm of dust/dirt on the outsole and still be able to close tightly, maintaining proper function.
2	Any mechanism in the sole must not negatively impact the comfort of the footwear.	The sole should not contain over 40% of any mechanism by weight%, with these being concentrated at the front (toe-guard) and rear (connector) of the shoe.
3	Upper must not compromise the performance benefits of the BiSole element.	The upper must securely be joined to both the connectors and the toe-guard so that the transfer of load through the shoe is exactly the same as a shoe with a typical sole.
4	The shoe's upper must be aesthetically pleasing and attractive to consumers.	In a survey I conduct, 60% or over must 'agree' or 'strongly agree' that my product is attractive or desirable.
5	The flipping mechanism must not interfere with walking or running motions.	The frontal mechanism's volume must be less than 5cm ² and be totally contained within the profile of the outsole, both from medial and lateral elevations.
6	The rotation and flipping mechanism must be resilient to harsh conditions as well as water and dust.	The mechanism must be corrosion resistant to maintain function after contact with water and enclosed to resist dust and dirt.
7	The closing/clip in mechanism at the rear must be secure and stable during any kind of use.	The clip in mechanism must hold the BiSole tightly throughout it's width, and allow for 2x a typical strike load without breaking or disconnection.
8	The clip-in mechanism must not interfere with walking or running motions of the user.	The volume of the back clip must be below 3cm ² and be totally contained within the profile of the outsole, both from medial and lateral elevations.

Sketch Analysis

Based on Requirements

To iterate further on each element of the BiSole, I separated the main components and looked further into the necessary changes and implementations that are required by my specifications.

In Detail: Mechanisms



Upper
The upper will be best using a pretty standard composition to appeal to users with some adaptations to better fit the more symmetric shape.

4

3

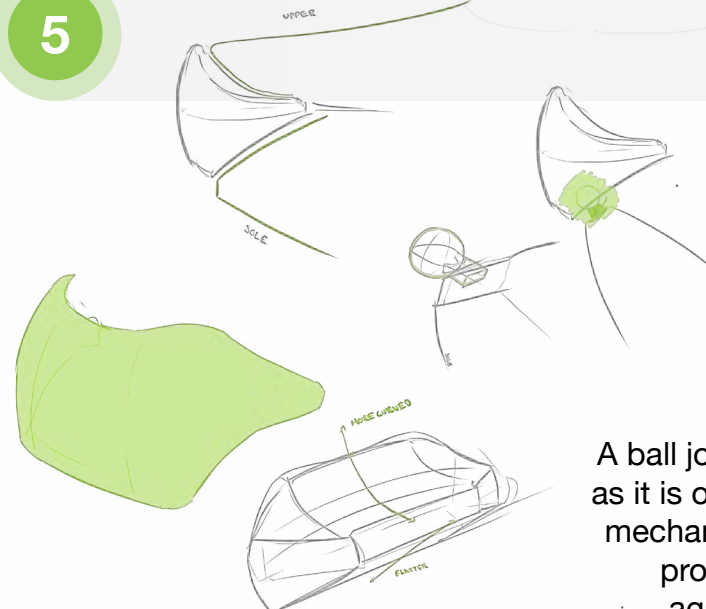
Upper Cont.

The upper of the shoe is likely to be made from a more breathable material rather than something solid to retain all of the athletic benefits of the BiSole.

Toe Cap

More detail in the toe-cap shows that the most compact mechanism would be a ball and socket joint. This allows normal range of movement in the toe.

5



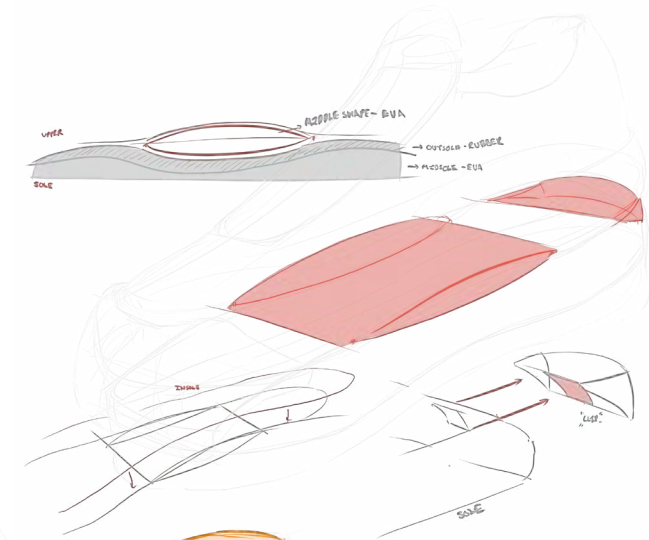
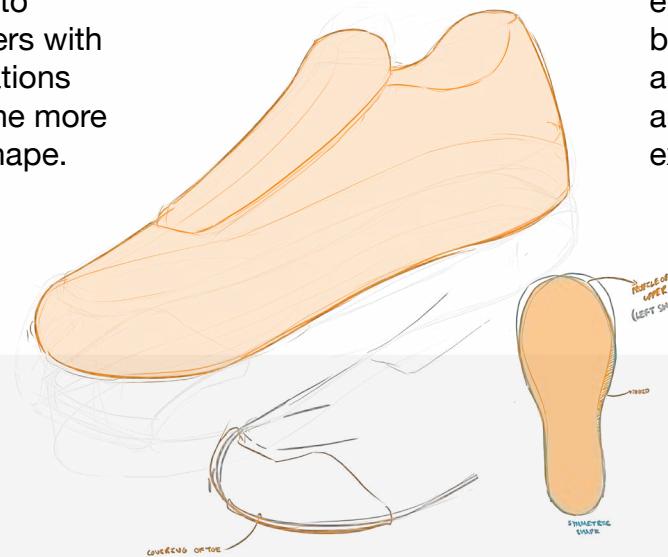
Toe Cap Cont.

A ball joint may be reliable as it is only a simple single mechanism but may have problems with sealing against dust and dirt.

6

Centre Connector
This could contain high friction or magnetic surface to ensure that the joint between the BiSole and Upper is tight and reliable during exercise.

7



8

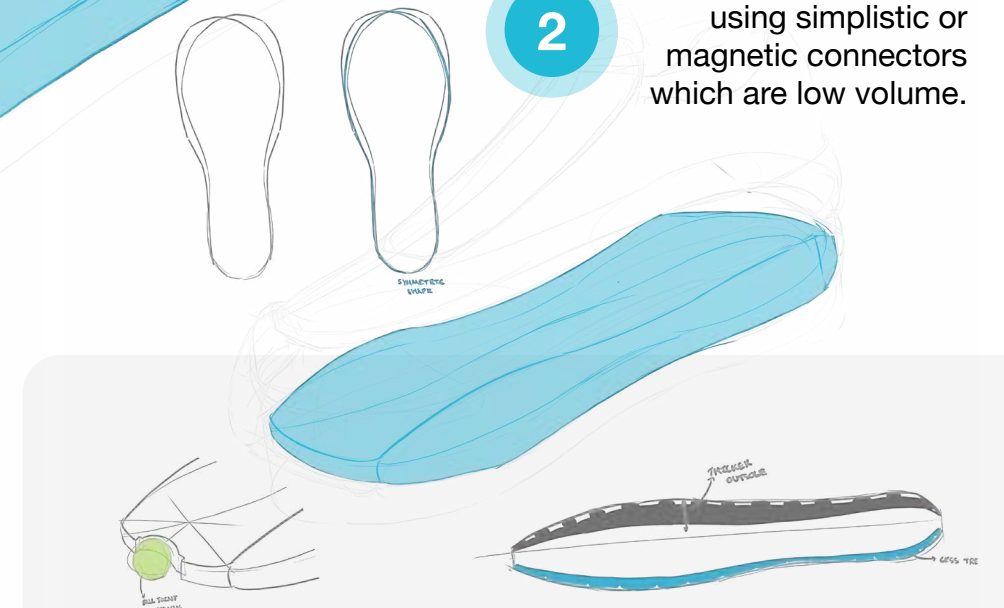
Rear Connector

The connecting mechanism that joins the BiSole and upper will be fully inside of the rear connector and could use a combination of magnets and mechanical fasteners to let it be small in volume.

BiSole Cont.

Flipping mechanism can be connected to the BiSole with a cutout, as well as the connectors using simplistic or magnetic connectors which are low volume.

2



BiSole

As well as a symmetric sole shape and varied outsole thickness, the BiSole element will need to have a certain degree of flexibility to accommodate any entrapped dust/dirt between the upper and BiSole.

1

Storyboard

Demonstrating the Function of the BiSole



Running on road terrain with **road outsole**.
Trail outsole is facing inwards.

1



Road terrain changes into the trails.
Road outsole is still facing outwards.

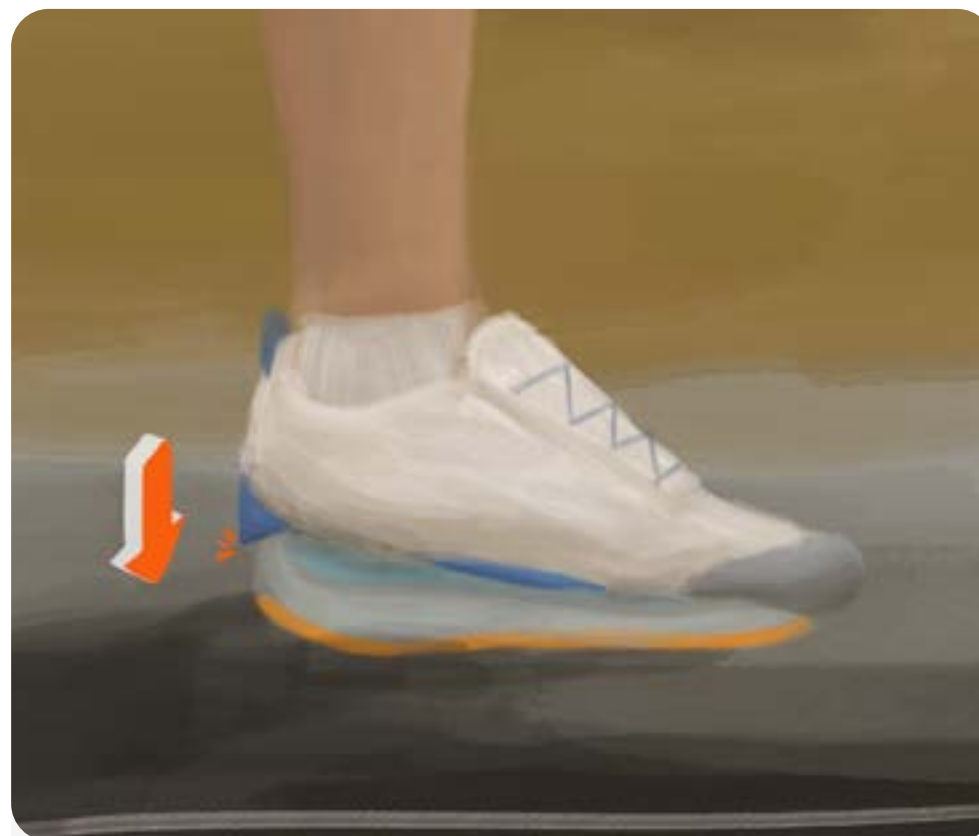
2

in order to demonstrate the function of the BiSole, I created a short storyboard which shows the simple process of swapping or 'flipping' the outsoles.



Unclip and flip the BiSole. Change from **road outsole** to **terrain outsole**.

3



"Click" the BiSole into place.
Terrain outsole is now facing out.

4



Running on the trails with **trail outsole**.
Road outsole is facing inwards.

5

Morphological Analysis

Materials and Mechanisms

Justification:

Upper: Mesh Construction
More structural than a knit but also more appropriate for a sports shoe than leather or Gore-Tex due to better breathability.

Toe Guard: Natural Rubber
Offers great protection and grip to the toe area as the guard extends over the outsole. Can be made thin to keep flexibility.

Flipping Mechanism: Inner Ball-Joint
Simple and reliable as previously explored in Sketch Analysis.

Centre 'Connector': EVA
Provides good cushioning and keeps complexity low in the central area of the shoe that is already complicated by the BiSole.

Heel 'Connector': PU
Higher density is better for area with large strike forces as well as allowing for magnets and a mechanism to be embedded inside of it.

Heel Locking Mechanism: Magnetic Clip
Using A magnet aided clip will ensure a secure fastening as well as allowing the overall volume of the mechanism to remain low.

Outsole Material: Vibram
Proven to be some of the best outsoles available for trail and hiking shoes due to all weather grip and low break-in periods.

BiSole Material: EVA
Abundant and readily available foam with great mix of compression and durability.

Selection:

Component	Option 1	Option 2	Option 3	Option 4
	Knit Very flexible and will be form fitting.	Leather Hard-wearing, stiff and weatherproof.	Gore-Tex Weather-proof but lacks breathability.	Mesh Very breathable but has lower flex.
Toe Guard	PVC Hard wearing synthetic rubber.	Rubber Durable and high level of grip.	Vibram High durability but stiff and inflexible.	Outsole-EVA Harder wearing form of EVA foam.
Flipping Mechanism	Outer Ball-Joint BiSole element slides over sphere.	Inner Ball-Joint BiSole moves inside of sphere.	Double Pivot One to swing out, other to spin sole.	Hinge and Clip Sole swings out, is unclipped + spun.
Centre 'Connector'	EVA Soft, light and well cushioned.	CFRP Plated Provides 'spring' or energy return.	PU Higher hardness, more durable foam.	PVC Highly durable and weather resitant.
Heel 'Connector'	(Outsole)-EVA Harder wearing form of EVA foam.	PVC Rubber Higher durability for strike forces.	E-TPU Mid-density foam, good cushioning.	PU High hardness but lower cushion.
Heel Locking Mechanism	Magnetic Clip Using magnets to attach the BiSole.	Sprung Catch Held in by spring and pin system.	Electronic Clip Sole fixed by motorised pin.	Friction Fit Two interlocking shapes that grip.
'Road' and 'Trail' Outsoles	PVC Hard wearing and Insulating.	Rubber High grip and good in the rain	Vibram Great traction and break-in time.	Outsole-EVA Low density and highly flexible.
'BiSole' Midsole	PU Excells in durability but lower comfort.	EVA Light, flexible and good support.	E-TPU Slightly higher density/cushioning.	PEBA Lightest/softest but least durable.

Bench- marking

Comparisons to Competitor Products



£120

SALOMON SENSE RIDE 5

A more typical performance trail running shoe designed for a variety of terrains and to appeal to a majority.

Advantages vs BiSole

Has a lighter and more 'responsive' foam in the midsole.

Simpler construction could make it more reliable or less likely to break.

Overall construction makes the shoe very light, which is great for a trail running applications.

Has a 'Quicklace' fastening system for easy on/off.

Dissadvantages vs Bisole

BiSole has the ability to adapt to different terrains, making it suitable to a greater range of different terrains and conditions.

The BiSole's adaptability also makes it safer in a variety of conditions, especially as the Sense-Ride is not recommended for wet weather.

The Salomon's construction also means it cannot be re-soled, which does not promote sustainability.



£200

SPEEDLAND GL:PDX

The successor to the SL:PDX trail shoe featured in my Initial Exploration, the GL version still has modular features.

Advantages vs BiSole

Has a combination of straps and BOA system to provide a consistent tight fit which is not possible with the BiSole.

Has removable 'Carbitex' plate which can provide more energy return or a greater level of stiffness to the insole.

The upper of the PDX has a combination of mesh and knit elements that allow for a more consistent fit.

Dissadvantages vs Bisole

The PDX's features cuttable sole blocks which can change to fit into the terrain but this is only single use and permanent.

The cuttable blocks also don't allow the SPEEDLAND shoe to change quickly or 'on the fly' like the BiSole can.

The BiSole is also likely not to be as expensive as the SPEEDLAND which also requires around £30 for the carbon plate features.



£230

NNORMAL KBOIX

A much more experimental piece of trail running footwear that's main feature is a swappable midsole.

Advantages vs BiSole

Has 3 choices of swappable midsole, allowing the KBOIX to be more customisable to the user.

It also allows the midsole to be changed if it is damaged, promoting sustainability.

This shoe also uses a mesh that is laced with carbon fibres to give a much more durable and long lasting material than in the BiSole.

Dissadvantages vs Bisole

BiSole is better at adapting to the terrain than the NNormal as it changes the part that physically interacts with the ground.

Similarly to the GL:PDX, the NNormal cannot be transformed quickly or in the place where it is effected by the terrain.

The NNormal shoe also has an extremely high barrier to entry, with the the shoe and one of each midsole type costing over £200.

body text all through this box and bot-
tom aligned if that is possible

Rankings:

After my comparisons I decided to rank each product on a scale of 1-10, including the BiSole to see how they all stack up against eachother.

9.1

8.5

7.2

5.0

Overall I decided on the NNormal as my benchmarking winner as, despite the high price, it provides the most innovative but realistic solution to the main problems within trail running.

Final Design

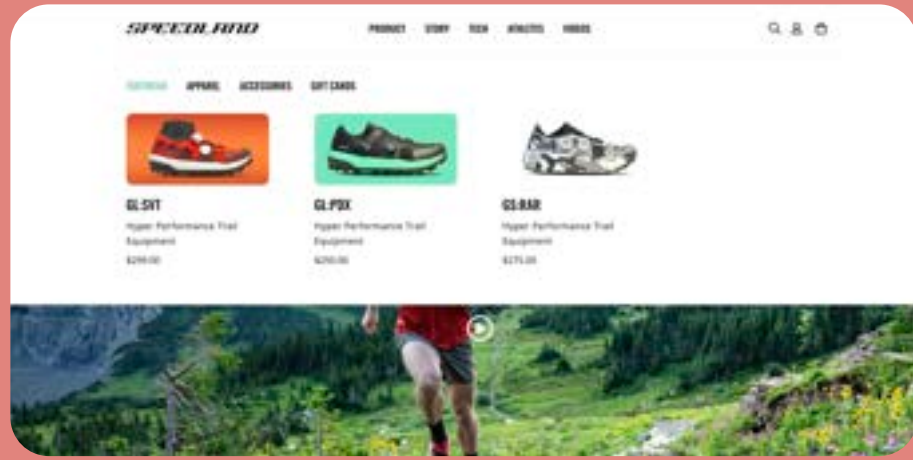
'BiSole' Footwear
Concept



BiSole

Design Validation

Product Analysis



Market Analysis:

Is there a market for the BiSole Concept?

Definitely, the current market in the trail running footwear space is competitive but there is room for innovation. Brands like SPEEDLAND and NNormal are showing that modularity and extreme performance are a great combination. It is also great to see that SPEEDLAND has expanded their range of footwear, clearly showing that there is demand.

Is it Possible to Justify a High Price Point?

£200

£230

Based on the current (small) market for innovations within the space, yes.

Higher prices are already very common for high performance footwear and this only seems to be compounded with modular or transformable elements.



Through market analysis, comparison against requirements and evaluation of the wider context, I aim to analyse the success of the BiSole concept.

Requirements:

Does The BiSole meet all of the requirements set out earlier in the project?

Yes, the final BiSole concept does meet all of the 8 requirements I established in my specification.

However, it is important to note that the majority of my specification points were physical, which means I was unable to test them on a final product.

Specifically looking at the requirements related to the upper of the shoe, although I was able to iterate in my sketch analysis page, I feel that there wasn't enough development in this area.

Focusing on the requirements that related to the sole elements of the shoe such as the connectors and BiSole, the overall product feels a lot more polished and clearly meets all of the quantitative specifications that I set out.

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Wider Context:

Does The BiSole concept help solve the environmental problem in footwear?

Simply, no. The BiSole concept cannot produce a large enough difference between it and traditional pairs of footwear. Only saving around 5% of production Carbon Dioxide Emissions over the lifetime of a pair of shoes.

However, the BiSole is also an effective performance concept, which means it is more geared to tackle the problem of overconsumption in athletics.

The adaptability of the BiSole is extremely important, with this being the reason it would be able to replace a runner's 'rotation' of shoes and therefore counteract the environmental damages of more pairs of shoes.

24Bn

Pairs of shoes produced globally each year.*

0.2kg

CO₂ Saved vs 2 traditional pairs of sports shoes.

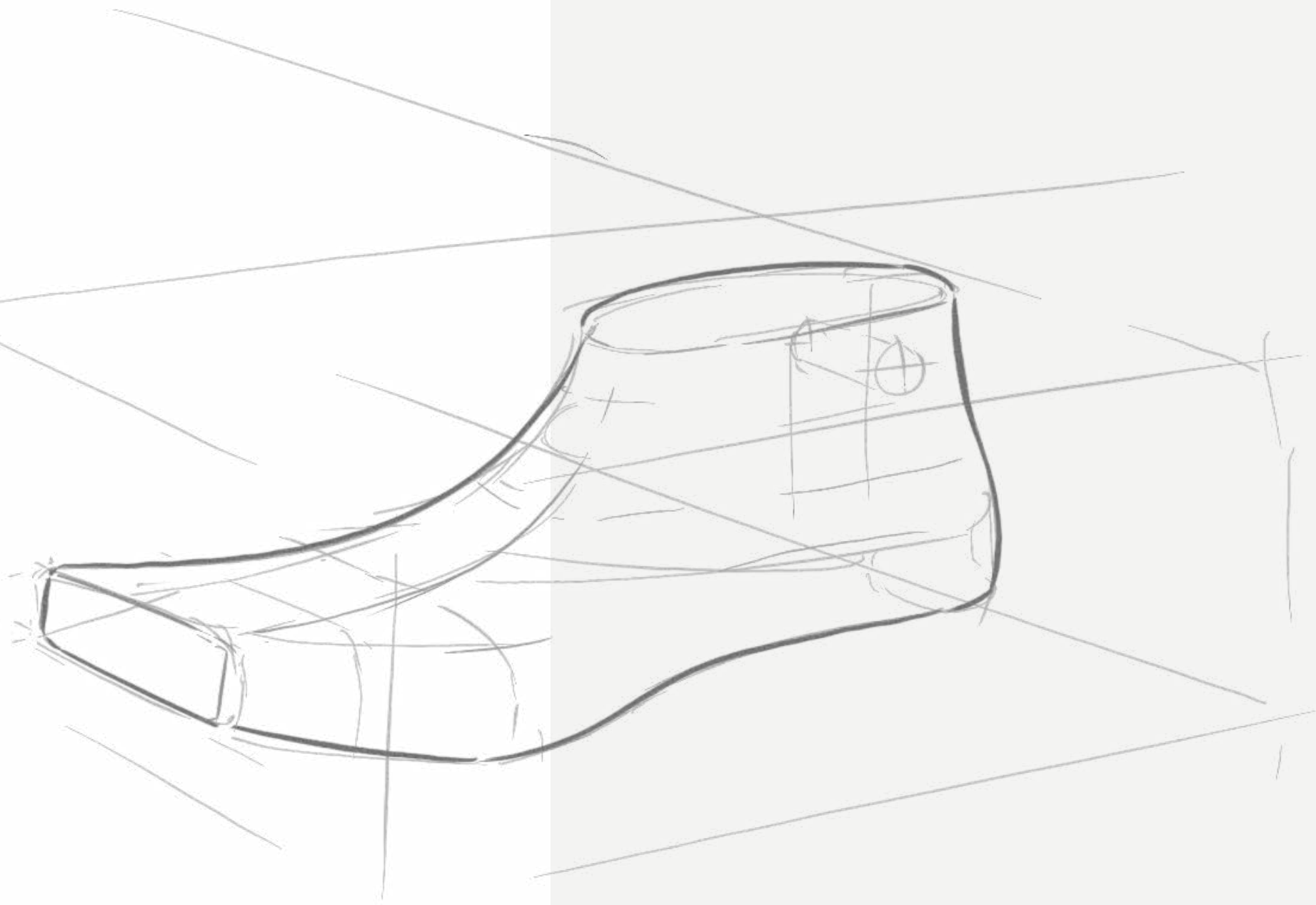
1.2M

Tonnes of waste created from footwear in Europe.***

~5%

Of CO₂ mass saved over BiSole lifetime.

Supplementary Pages



GREY BOXES
For most text although some parts didn't need it, the left side grey box is 1/3 of the page horizontally, but it changes size to fit the content of the page E.G 15 Ideas Analysis page.

All boxes have 5mm corner radii and are spaced with 5mm gaps.

#568CC7

#D65536

#252626

#178136

#F8F7F7

#37B4D5

#9CD436

#DC645B

#F7880D

InDesign Elements

Supplementary page 1.

Body/descriptive text is bottom aligned and justified to fill out the bottom of the title box uniformly.

TITLE 66pt

SUBTITLE 35pt

Headings 17pt
Body 14pt
General Use 12pt

SUPPLEMENTARY

PRESENT

DEVELOPMENT

RESEARCH

2X3 GRID
WITH 5MM GUTTER
STANDARD ALTHOUGH
SOME PAGES HAVE
ADDITIONAL GRIDS FOR
MORE ELEMENTS.

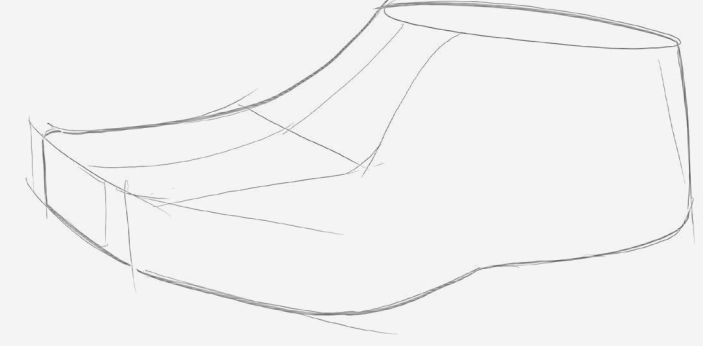
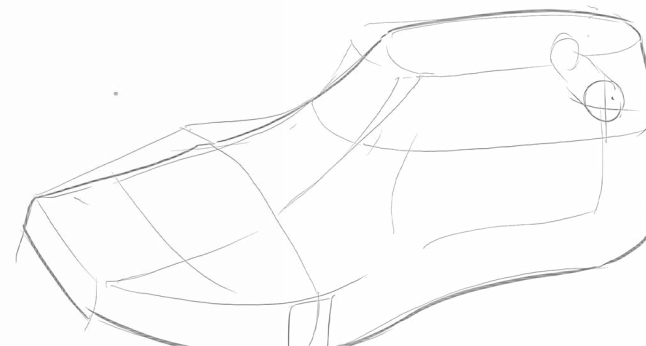
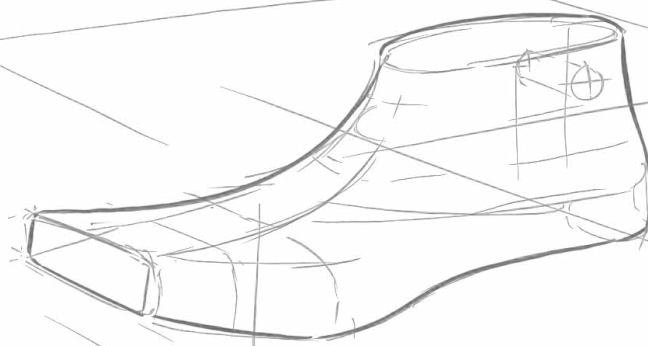
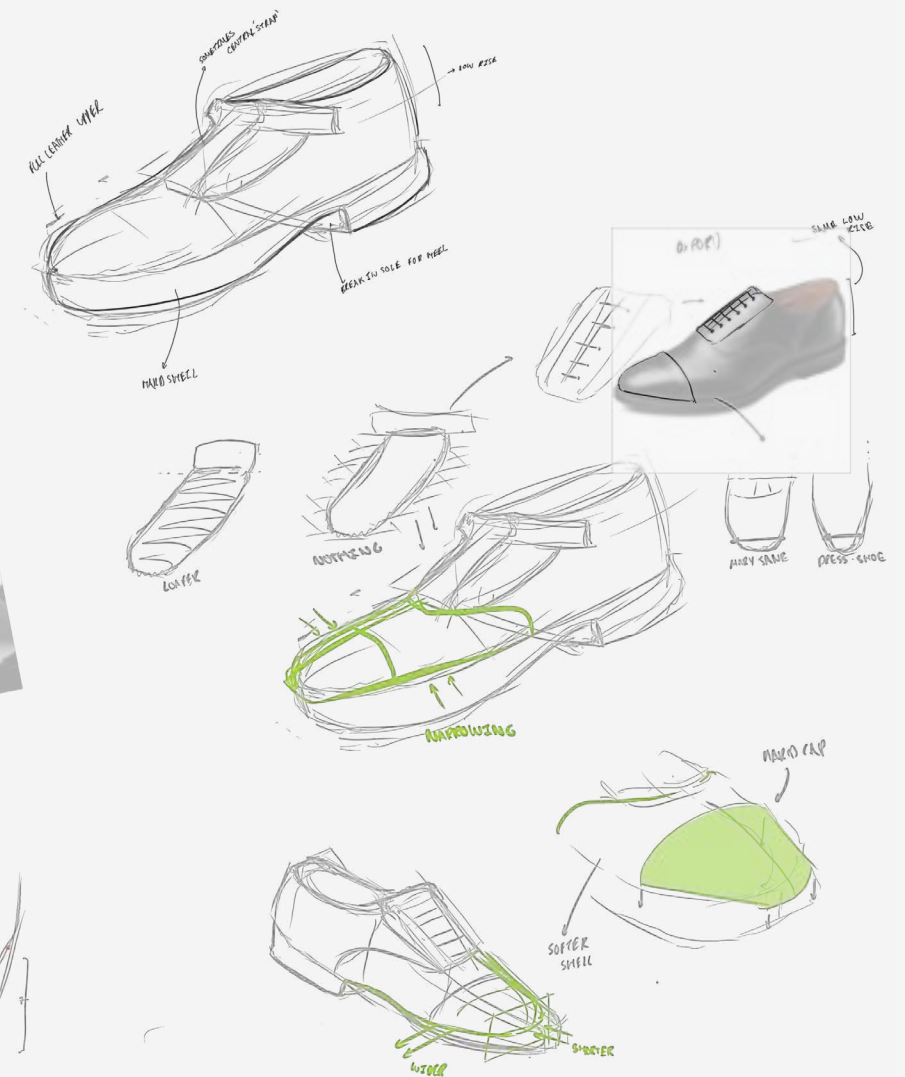
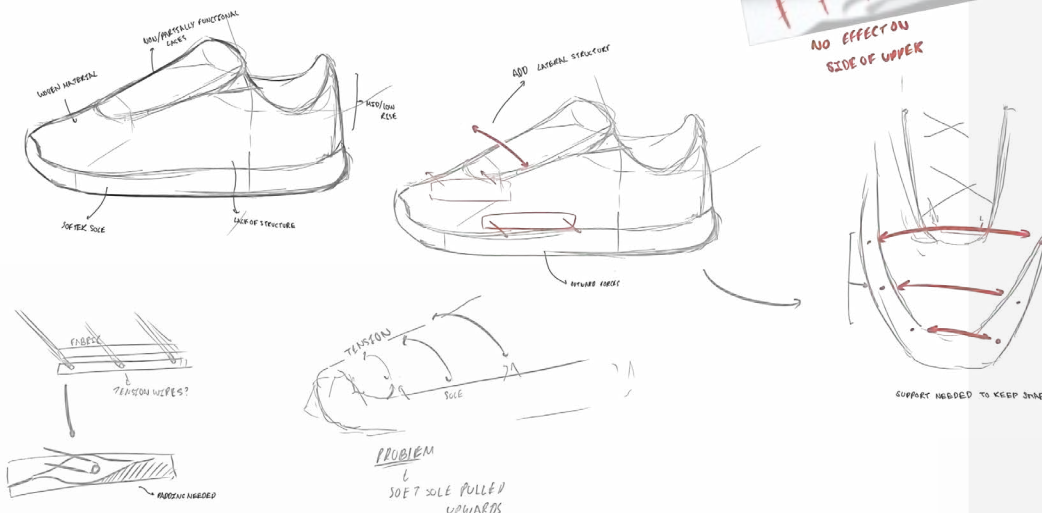
Sketching Practice

Supplementary page 2.

I practiced sketching all the way through the project: the top image shown is some initial sketches that I did around the same time as I was selecting my initial context.

Lower Images

These Sketches were done later in the project, before I started my final concept, they helped me to get a better understanding of the proportion by drawing the shoe-last.



How Might We (HMWs)

Supplementary page 4.

After establishing Insights through 5W&1H questions, I was able to convert some of these into How Might We Questions. Starting by directly converting insights into HMW I then worked to combine and simplify into one main How Might We question using AI tools like OpenAI's ChatGPT.

HMW Design **Sports** Footwear to Transform for Different **Terrains** and Activities?

Generating HMWs

Initial HMW questions from Analysis.

HMW enable footwear to adapt to terrain?

HMW allow footwear to change in response to its environment?

HMW adapt the properties of shoes to better suit a high performance user?

HMW design shoes that change to ensure safety in dangerous terrains?

HMW better allow sports footwear to adapt to changes in terrain?

Refining HMWs

Using ChatGPT to combine and enhance the specifics of my HMWs.

HMW enable athletic footwear to adapt its materials to different terrains for enhanced performance and safety?

HMW rethink footwear for better performance across diverse environments?

HMW design athletic footwear that dynamically adjusts its grip to optimize performance across various sports surfaces?

HMW explore technologies and designs that allow athletic footwear to seamlessly adjust to changing conditions and activities?

HMW Better Design Sports Footwear to Dynamically Transform for Different Terrains and Activities?

HMW enable shoes to seamlessly transition between different uses, such as hiking, running, and climbing, by reconfiguring their design?

HMW enable athletic footwear to respond in real time to changes in terrain, such as track, grass, or court?

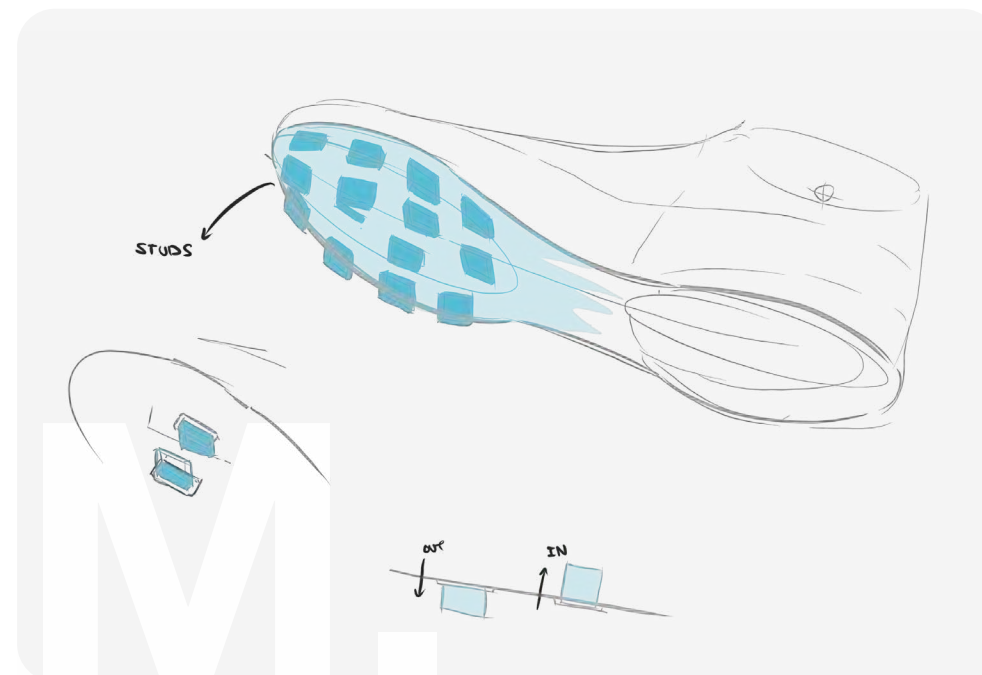
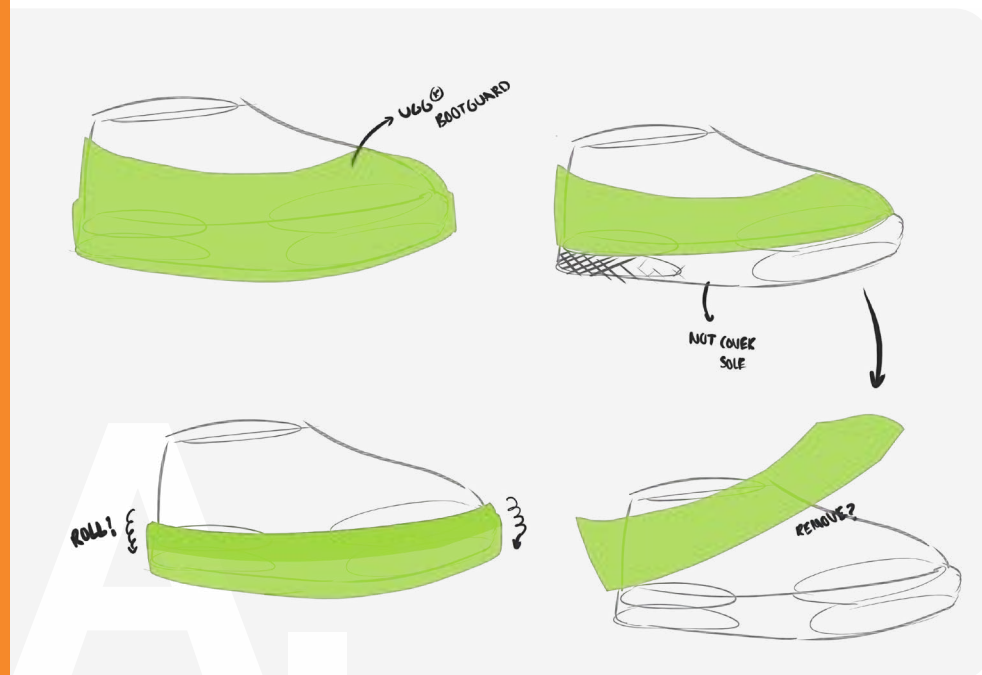
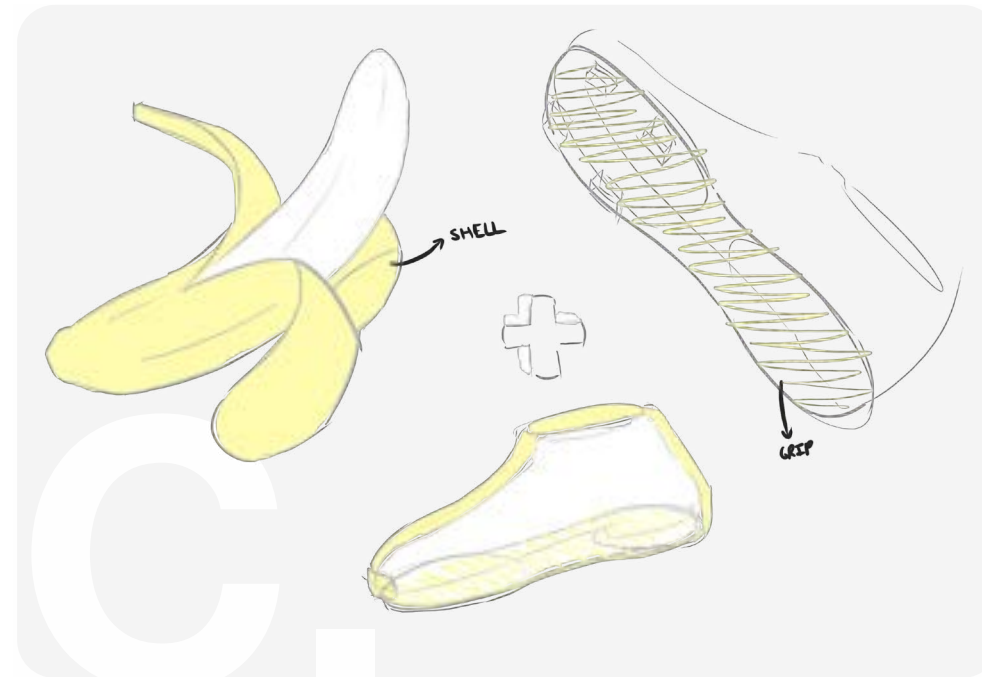
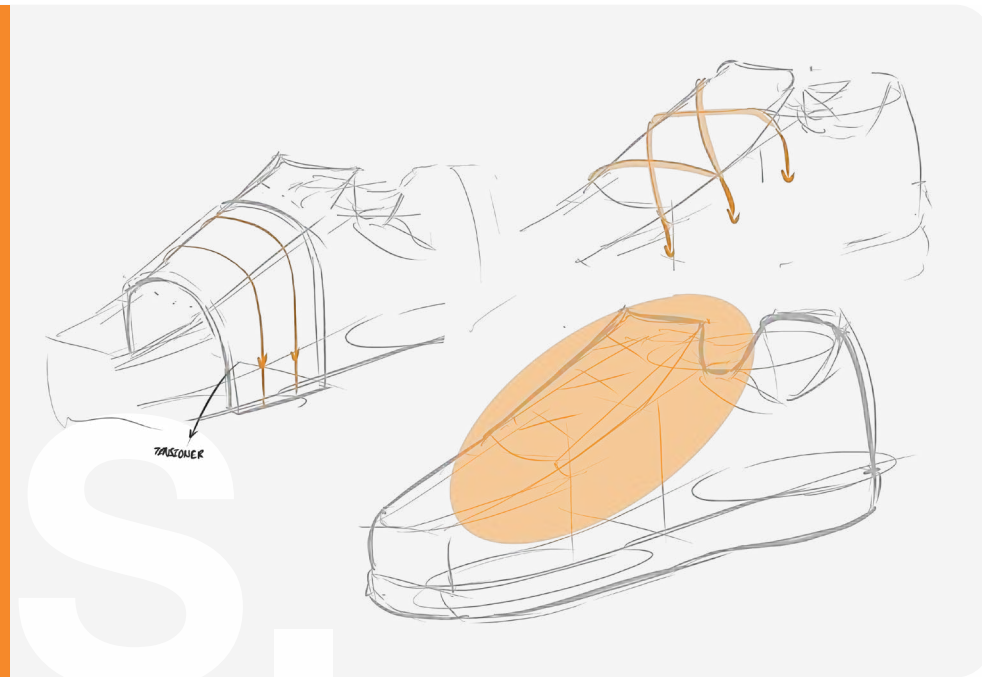
HMW design footwear that adjusts to the demands of various surfaces?

HMW create shoes that dynamically transform their structure and support to enhance performance and safety in any environment?

HMW explore new material combinations that allow footwear to change shape, texture, or support based on terrain needs?

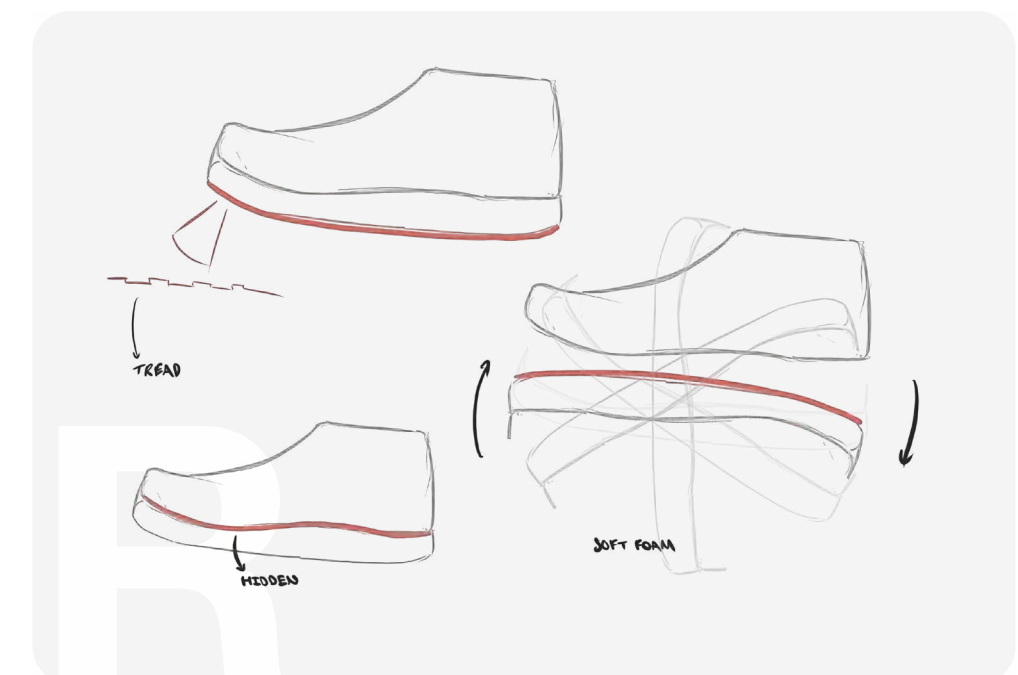
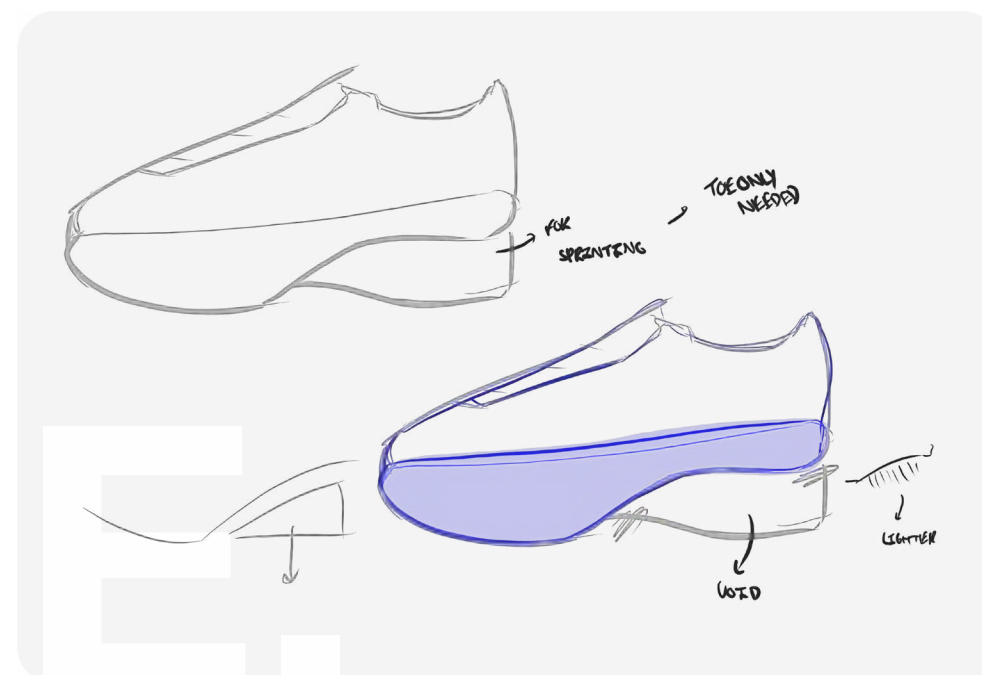
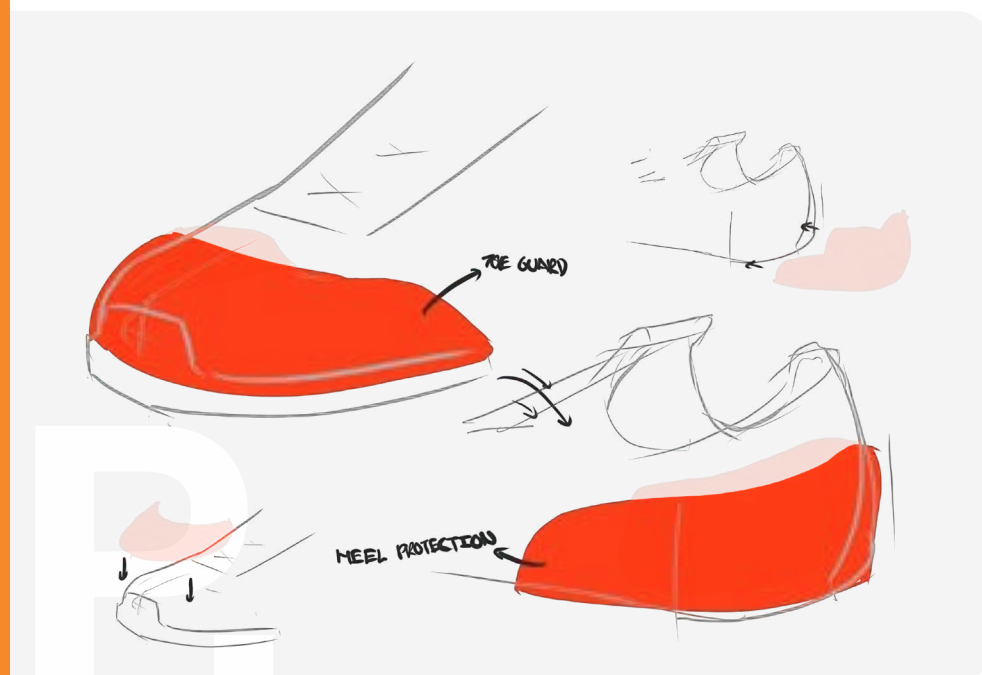
Secondary Ideation

Supplementary page 5.



Moving forward from my initial ideation, I used the S.C.A.M.P.E.R method to start creating unique ideas. Basing a lot of these concepts on ideas explored through my first sketches as well as working within my HMW helped me make my ideas relevant and unique.

Substitute. Combine. Adapt. Modify. Put-to-Another-Use. Eliminate. Reverse.



Idea Selection

Supplementary page 6.

To select the concepts presented in my 15 ideas section, I went back to my initial analysis and elaborated on a lot of the first designs to make them suitable for presentation.



Additional Prototypes

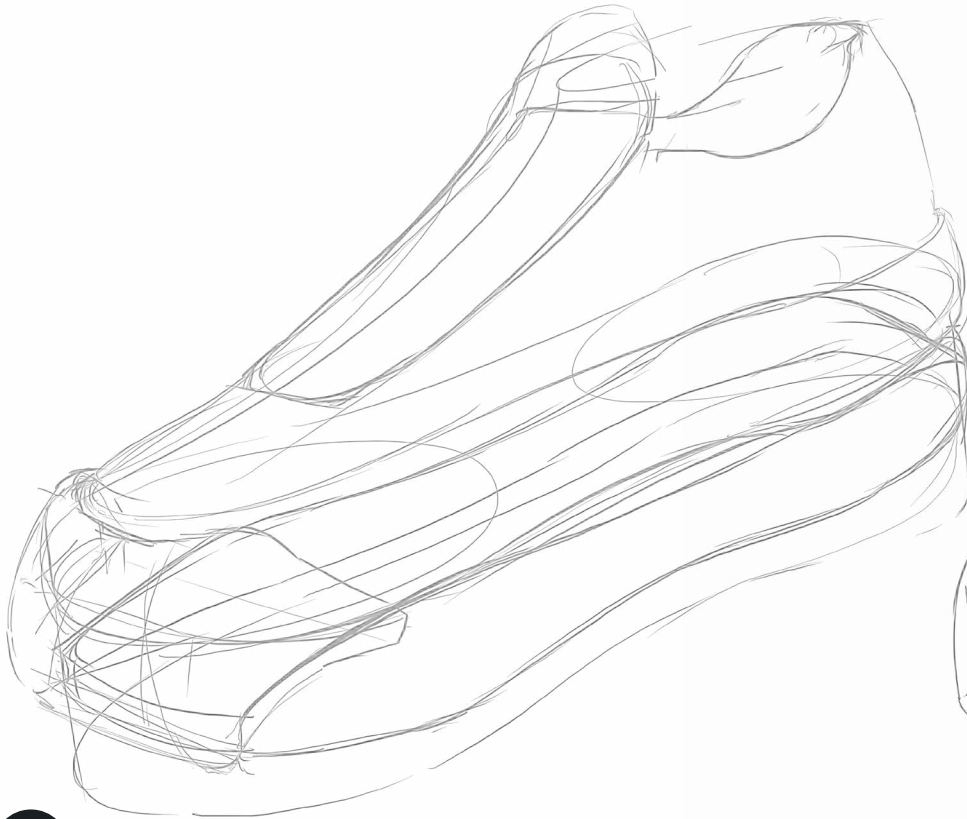
Supplementary page
7.

More images from my prototyping process, some before I started the lo-fi prototypes of my 3 selected concepts.

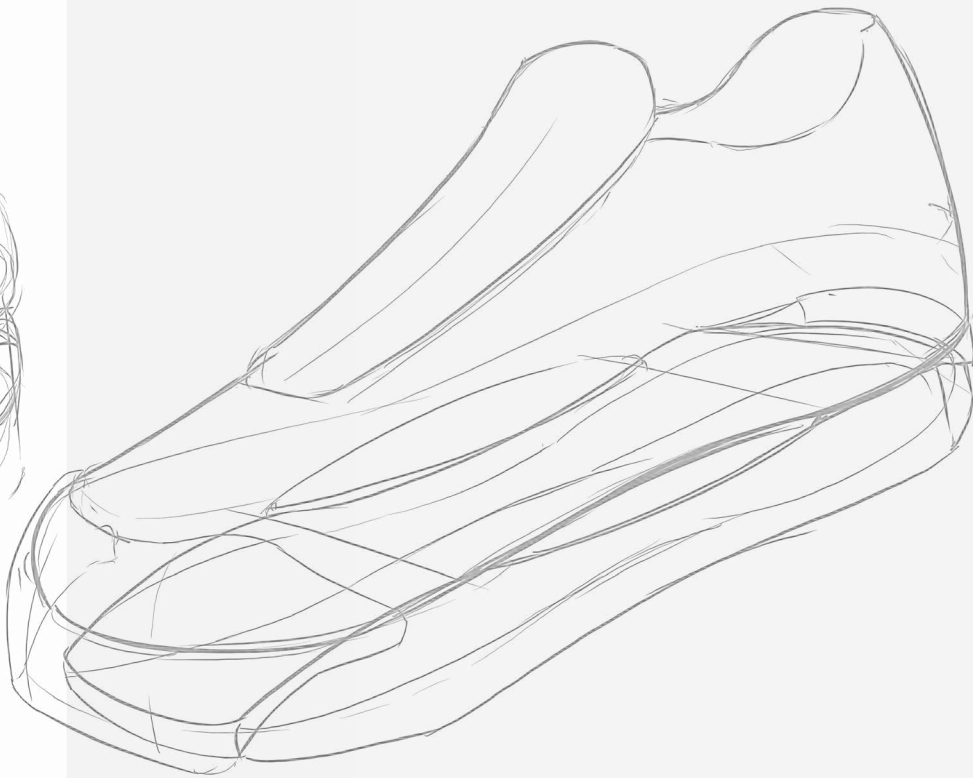
Concept Sketching

Supplementary page 8.

Showing the sketching process for my 'Single Concept' drawings on page 11.



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AI Concept Renders

Supplementary page 9.



Taking my Single concept sketches and using Viscom AI to create a realistic render allowed me to see areas of the shoe that didn't translate well into realism.



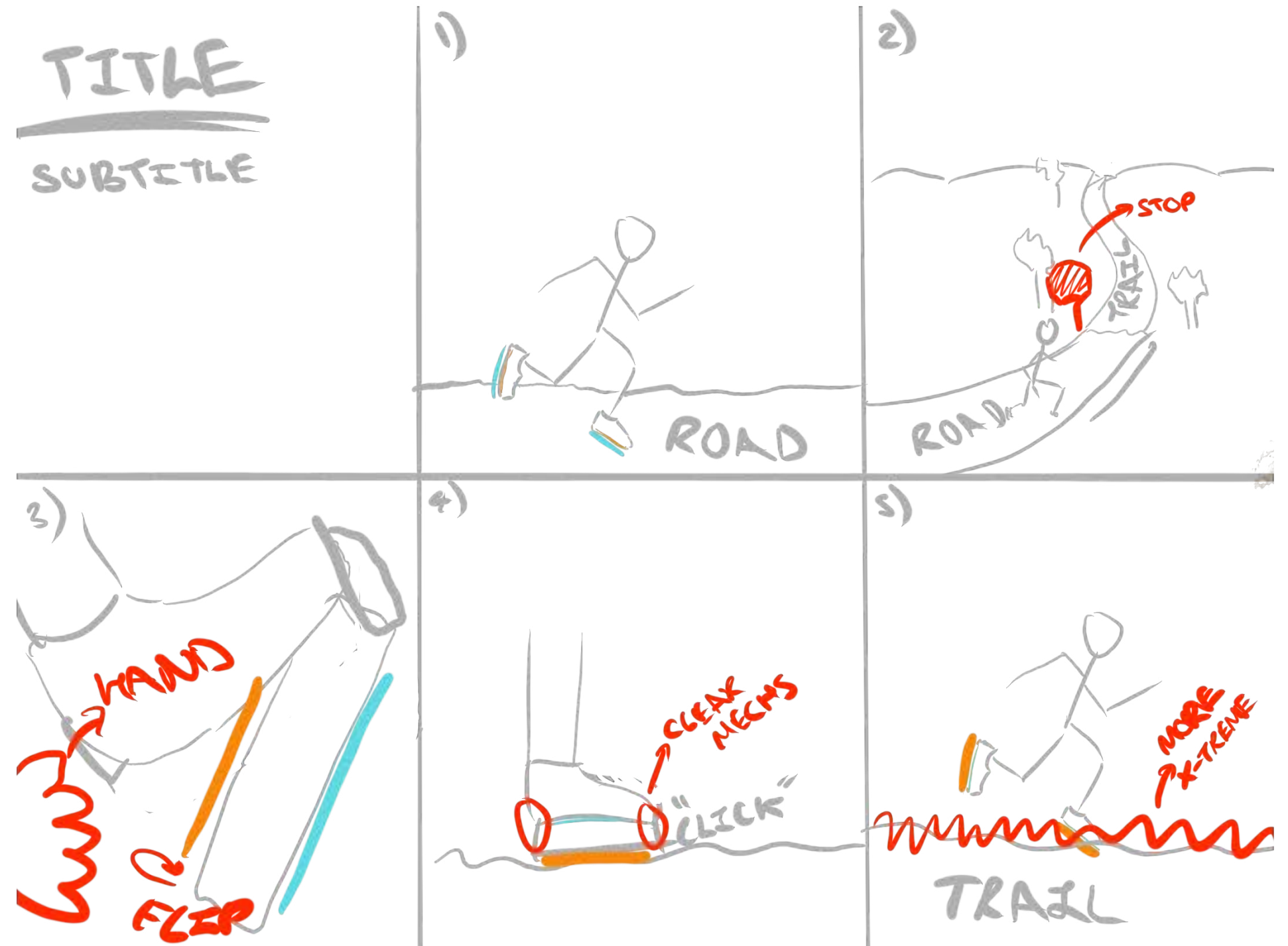
Images Generated by Viscom AI

Storyboard Planning

Supplementary page 10.

My initial and elaborated plan for my storyboarding page where the functional steps of the BiSole are demonstrated.

-  Original sketch
-  Road Outsole
-  Trail Outsole
-  Review and add



Grey Drawings

These are my initial drawings, after I decided that a short 5 slide storyboard would be enough to explain the shoe's function.

Red Drawings

After my initial drawings I consulted some peers on how I could make the storyboard more clear.

CALCULATION OF CO₂ FOR BISOLE

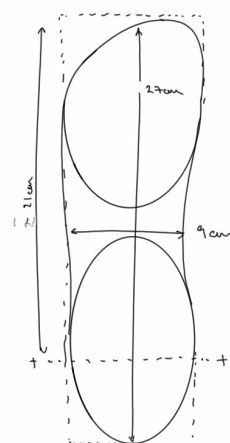
BISOLE

EVA

$$\rho = 60 \text{ kg/m}^3$$

$$\text{CO}_2 \text{ emissions} = 2.2 \text{ kg CO}_2/\text{kg}$$

$$\text{Volume} = 21 \times 9 \times 3 \times (10^{-2})^3 = 0.000567 \text{ m}^3$$



$$V = 0.000567 \text{ m}^3$$

$$\rho = \frac{m}{V} \quad m = \rho V$$

$$\rho V = 0.000567 \times 60 = 0.03402 \text{ kg}$$

$$\text{per shoe} \therefore \text{kg CO}_2 = 0.074844 \text{ kg}$$

$$\text{per pair} = 0.149688 \text{ kg}$$

'VIBRAM' Outsoles

CO₂

$$\text{release} \approx 1.94 \text{ kg} \\ \text{per pair of} \\ \text{outsoles}$$

BiSole uses 2 outsoles
per shoe

So for pair

$$= 1.94 \times 2 = 3.88 \text{ kg}$$

$$\text{Total} = 4.02988 \text{ kg}$$

$$\approx 4.03 \text{ kg of CO}_2 \\ \text{emitted.}$$

However With use of 'I'm Green'

EVA, emissions drop to around

$$3.90 \text{ kg}$$

COMPARISON CALCULATION

EVA

2.5 cm

Some S.A

$$\therefore V = 0.0004725 \text{ m}^3$$

$$m = 0.02835$$

$$\text{kg of CO}_2 = 0.06237$$

$$\text{per pair} = 0.1274$$

$$\text{Total} = 0.1274 + 1.94 \text{ kg} \\ = 2.0674$$

$$2.0474 \\ \text{per pair of} \\ \text{Normal shoes}$$

Calculations

Supplemental page 11.

To compare the environmental Impact of the BiSole to a usual pair of shoes, I calculated the CO₂ Emmisions of the Soles of both a normal pair and the BiSole Concept.

4.08kg

Per Pair of BiSole Shoes.

2.05kg

Per Pair of Normal Shoes.

Assumptions

My main assumption before these calculations is that the upper portion of the shoes can be considered to be irrelevant/negligible. I decided to do this to focus on the sole region as it is the greatest physical difference between the BiSole and a traditional shoe.

Despite this, the uppers of the shoes would be calculated to be the same, which would only increase the environmental benefit of the BiSole.

References

Vibram Outsole CO₂ Emmisions: (No date) The sustainable way. Available at: https://eu.vibram.com/on/demandware.static/-/Sites-VibramEU-Library/default/dw8032ef79/pdfs/Vibram_The_Sustainable_Way-Summary_of_2021_Sustainability_report_ENG.pdf (Accessed: 04 December 2024).

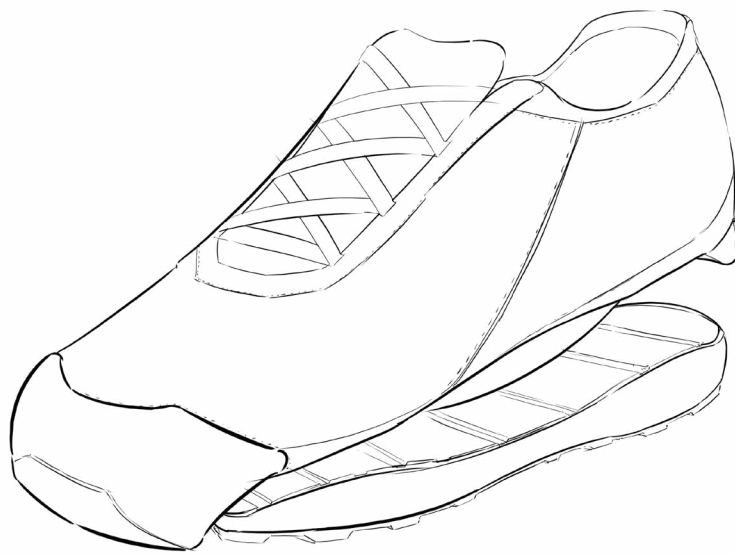
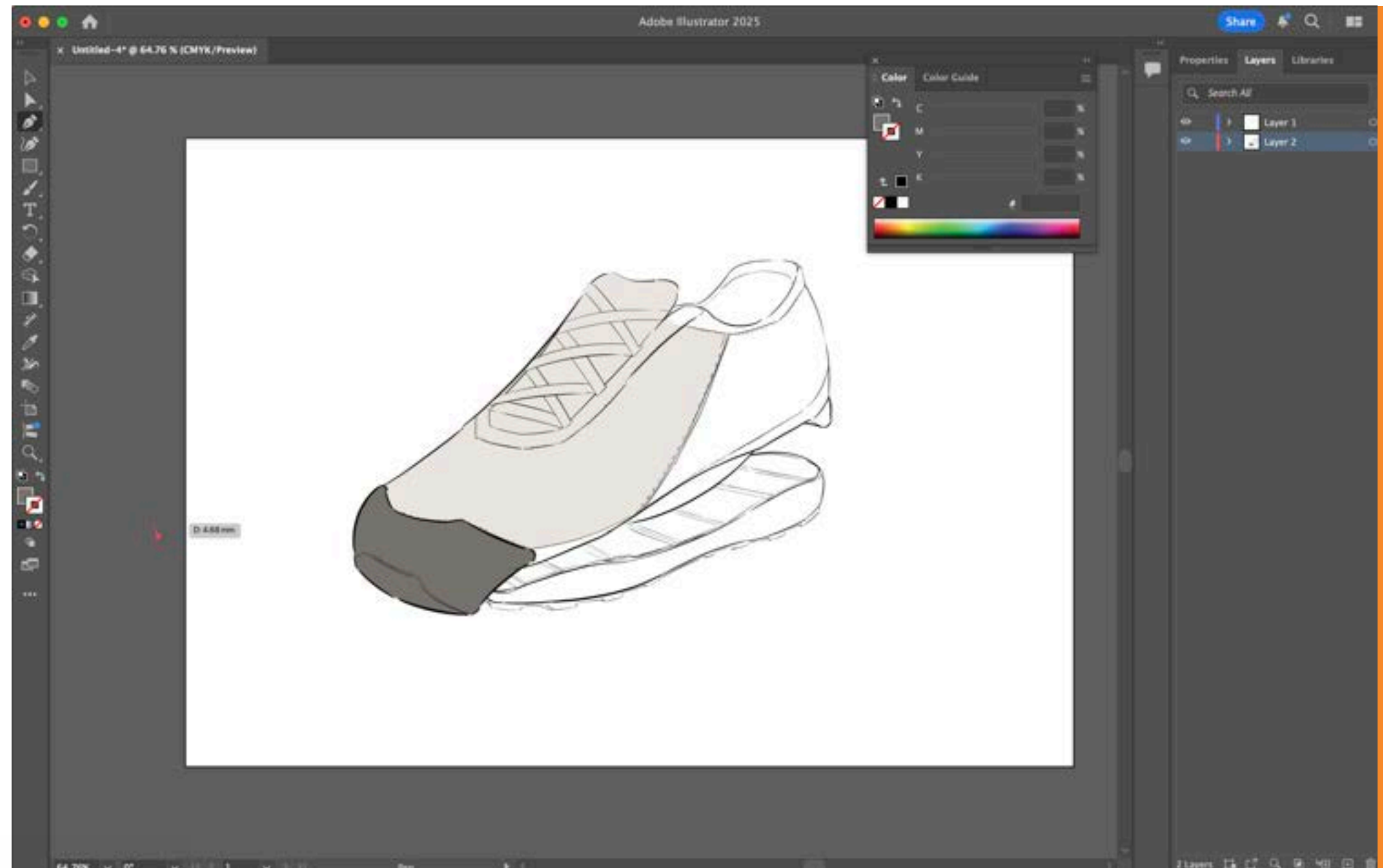
EVA Foam Density: (No date) The sustainable way. Available at: https://eu.vibram.com/on/demandware.static/-/Sites-VibramEU-Library/default/dw8032ef79/pdfs/Vibram_The_Sustainable_Way-Summary_of_2021_Sustainability_report_ENG.pdf (Accessed: 04 December 2024).

EVA CO₂ per kg: (No date a) Useful numbers for environmental studies. Available at: <https://www.polybags.co.uk/environmentally-friendly/useful-numbers-for-environmental-studies.pdf> (Accessed: 04 December 2024).

Adobe Illustrator

Supplementary page 12.

To experiment with adobe illustrator, I decided to create final design images in the software for the portfolio.



Starting by creating a sketch and then a line drawing of the BiSole in Procreate, I then took that into illustrator, using the pen tool and 'create sketch' features, Was able to vectorise my sketch. After this I used different blocks of colour to give the sketch colour and shading.



IMAGES

PAGE 2

Graph:Rahimifard S, Staikos T, Coates G (2007) Recycling of Footwear Products.
Adidas Adifom Q: Unknown, found at <https://www.kickgame.co.uk/products/adidas-adifom-q-off-white>
Salomon Snowclog advanced: Unknown, found at <https://www.salomon.com/en-gb/shop-emea/product/snowclog-advanced-li6173.html#color=103021>
SPEEDLAND SL:PDX: Unknown, found at <https://believeintherun.com/shoe-reviews/speedland-slpdx-performance-review/>
Crocs: Unknown, found at <https://www.crocs.com.sg/p/everything-nice-5-pack/10008657.html>

PAGE 15

SALOMON SENSE RIDE. Ssource: <https://www.salomon.com/en-gb/shop-emea/product/sense-ride-5-li3121.html#color=83560>
SPEEDLAND GL:PDX. Ssource: <https://www.runspeedland.com/products/gl-pdx>
NNORMAL KBOIX. Ssource: https://www.nnormal.com/en_GB/men/shoes/tomir/nnormal-kboix-N2ZKXM1-999-K001

CREDITS

Shoutout to all the people who helped me consistently through the project your peer review insight was always useful. :)

STATISTICS

PAGE 2

*23.9Bn pairs produced in 2022. Source: Statista Research Department (2024) Amount of footwear produced worldwide from 2015 to 2022. Referencing World Footwear Yearbook 2023.
**Source: Van Rensburg ML, Nkomo SL, Mkhize NM. Life cycle and End-of-Life management options in the footwear industry: A review. Waste Management & Research. 2020;38(6):599-613. doi:10.1177/0734242X20908938
***Source: European Commision LIFE public database. Demonstration of a new business and consumption model for the circular economy in the footwear sector. LIFE19 ENV/ ES/000118
****Source: Fiona Armstrong-Gibbs UK Parliament written evidence. Evidence on Sustainability of the fashion industry: the footwear perspective. SFI002

PAGE 3

Recommended pairs: <https://www.i-run.com/running/news/how-many-running-shoes-should-i-have.html#:~:text=Optimal%3A%20four%20pairs%2C%20tailored%20to,the%20durability%20of%20each%20pair.>

PAGE 15

Price and features of SALOMON SENSE RIDE. Ssource: <https://www.salomon.com/en-gb/shop-emea/product/sense-ride-5-li3121.html#color=83560>
Price (GBP Converted) and features of SPEEDLAND GL:PDX. Ssource: <https://www.runspeedland.com/products/gl-pdx>
Price (GBP Converted) and features of NNORMAL KBOIX. Ssource: https://www.nnormal.com/en_GB/men/shoes/tomir/nnormal-kboix-N2ZKXM1-999-K001

Page 29

Vibram Outsole CO₂ Emmisions: (No date) The sustainable way. Available at: https://eu.vibram.com/on/demandware.static/-/Sites-VibramEU-Library/default/dw8032ef79/pdfs/Vibram_The_Sustainable_Way-Summary_of_2021_Sustainability_report_ENG.pdf (Accessed: 04 December 2024).
EVA Foam Density: (No date) The sustainable way. Available at: https://eu.vibram.com/on/demandware.static/-/Sites-VibramEU-Library/default/dw8032ef79/pdfs/Vibram_The_Sustainable_Way-Summary_of_2021_Sustainability_report_ENG.pdf (Accessed: 04 December 2024).
EVA CO₂ per kg: (No date a) Useful numbers for environmental studies. Available at: <https://www.polybags.co.uk/environmentally-friendly/useful-numbers-for-environmental-studies.pdf> (Accessed: 04 December 2024).

Credits and
References
Supplementary page
13.

All citations for images and facts as well as an account of the software used throughout the project.

